HYODO-KATO COHOMOLOGY IN RIGID GEOMETRY: SOME FOUNDATIONAL RESULTS

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Abstract. By exploring the geometric properties of Hyodo-Kato cohomology in rigid geometry, we establish several foundational results, including the semistable conjecture for étale cohomology of almost proper rigid analytic varieties and the GAGA comparison between algebraic and analytic Hyodo-Kato cohomology. A central component of our approach is the Gysin sequence for Hyodo-Kato cohomology, which we construct using the open-closed exact sequence for compactly supported Hyodo-Kato cohomology and Poincaré duality.

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1. Introduction

This article is devoted to proving geometric properties of Hyodo-Kato cohomology in rigid geometry. We establish the Mayer-Vietoris property, Poincaré duality, and the Gysin isomorphism for the Hyodo-Kato cohomology of dagger varieties. Using these properties, we prove a comparison between algebraic and analytic Hyodo-Kato cohomology (GAGA) and the semistable conjecture for étale cohomology of almost proper rigid analytic varieties.

Let $\mathscr{O}_K$ be a complete discrete valuation ring with fraction field K of characteristic 0 and with perfect residue field k of characteristic p. Let $W(k)$ be the ring of Witt vectors of k with fraction field F (therefore $W(k) = \mathscr{O}_F$). Let $\bar{K}$ be an algebraic closure of K and C be its p-adic completion, and let $\mathscr{O}_{\bar{K}}$ denote the integral closure of $\mathscr{O}_K$ in $\bar{K}$. Let $\bar{k}$ be the residue field of $\mathscr{O}_{\bar{K}}$, and let $W(\bar{k})$ be its ring of Witt vectors with fraction field $\breve{F}$ (thus $\breve{F}$ is the p-adic completion of F^{nr}, the maximal unramified extension of F in $\bar{K}$, and $W(\bar{k}) = \mathscr{O}_{\breve{F}}$) and let ϕ be the absolute Frobenius on $W(\bar{k})$. Set $\mathscr{G}_K = \mathrm{Gal}(\bar{K}/K)$.

We will denote by $\mathcal{O}_K, \mathcal{O}_K^{\times}, \mathcal{O}_K^0$, depending on the context, the scheme $\mathrm{Spec}(\mathcal{O}_K)$ or the formal scheme $\mathrm{Spf}(\mathcal{O}_K)$ with the trivial log structure, the canonical (i.e., associated to the closed point) log structure, and the log structure induced by $\mathbb{N} \to \mathcal{O}_K, 1 \mapsto 0$, respectively.

1.1. **Semistable conjecture for open varieties.** Hyodo-Kato cohomology for algebraic varieties first appeared in the Fontaine-Jannsen conjecture ([**?**] and [**?**]), also known as the semistable conjecture. This conjecture suggests the existence of a "new crystalline cohomology group" H^i_{HK}, which should compare with étale cohomology for a proper scheme $X/\mathcal{O}_K$ with semistable reduction.

Moreover, this group should serve as a deformation of de Rham cohomology, i.e., there should exist a Hyodo-Kato morphism $\iota_{\mathrm{HK}} : H^i_{\mathrm{HK}}(X_K) \to H^i_{\mathrm{dR}}(X_K)$ such that the base change

$$\iota_{\mathrm{HK}} \otimes_F K : H^i_{\mathrm{HK}}(X_K) \otimes_F K \xrightarrow{\simeq} H^i_{\mathrm{dR}}(X_K)$$

is an isomorphism. Moreover, the group $H^i_{\mathrm{HK}}(X_K)$ is endowed with a Frobenius ϕ and a monodromy operator N. The conjecture can be formulated as follows.

Conjecture 1.1. *Let X be a proper and smooth algebraic variety over K admitting a semistable model over $\mathscr{O}_K$. Let $i \geq 0$. We have a functorial $\mathscr{G}_K$-equivariant B_{st}-linear isomorphism commuting with φ and N*

$$H^i_{\text{ét}}(X_C, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \simeq H^i_{\mathrm{HK}}(X) \otimes_F B_{\mathrm{st}},$$

compatible with the de Rham period morphism, via the natural injection $B_{\mathrm{st}} \subset B_{\mathrm{dR}}$, and the Hyodo-Kato morphism $\iota_{\mathrm{HK}} : H^i_{\mathrm{HK}} \to H^i_{\mathrm{dR}}$.

Hyodo-Kato cohomology was defined in [**?**] for varieties with semistable reduction, by introducing a modified de Rham-Witt complex. Later in [**?**], Osamu Hyodo and Kazuya Kato introduced log-crystalline cohomology and used it to give another construction for varieties with log-smooth models. A major advance appears in [**?**], where Beilinson constructed the Hyodo-Kato cohomology for algebraic varieties by h-descent [**?**], and proved the semistable conjecture without any smoothness, properness, or reduction hypothesis.

For proper smooth rigid analytic varieties, the above conjecture (without the semistable reduction hypothesis) also holds by [**?**]. The proof relies on the geometrization of syntomic cohomology and the theory of Banach-Colmez spaces.

As in the algebraic case, it is natural to ask what happens for varieties that are not proper or not smooth. The situation is more complicated, especially if one wants to recover the étale cohomology. For pro-étale cohomology, various results have been obtained in [**?**], where Pierre Colmez, Gabriel Dospinescu, and Wiesława Nizioł calculated the p-adic étale cohomology and pro-étale cohomology for the p-adic Drinfeld half-plane. In the pro-étale case, their results, combined with [**?**], show that for $r \geq 0$ and a smooth Stein space X over C, we have the following commutative Galois-equivariant diagram, which can be regarded as a pro-étale version of the semistable conjecture:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \Omega^{r-1}(X)/\ker d & \longrightarrow & H^r_{\text{proét}}(X, \mathbb{Q}_p(r)) & \longrightarrow & \left(H^r_{\mathrm{HK},\breve{F}}(X) \widehat{\otimes}_{\breve{F}} B_{\mathrm{st}}\right)^{N=0,\varphi=p^r} & \longrightarrow & 0 \\
 & & \| & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \Omega^{r-1}(X)/\ker d & \longrightarrow & \Omega^r(X)^{d=0} & \longrightarrow & H^r_{\mathrm{dR}}(X) & \longrightarrow & 0.
\end{array}$$

In [**?**], there is also a semistable conjecture which generalizes the above result for arbitrary quasi-compact dagger varieties. However, this conjecture remains open.

In this article, we consider the étale version of the semistable conjecture for open varieties. We show that the semistable conjecture is true for almost proper rigid analytic varieties.

Theorem 1.2. *(Theorem **??**) Suppose X is a proper smooth rigid analytic variety over C, $Z \subset X$ is a strict normal crossings divisor, and $U = X - Z$.*

(1) We have a B_{st}-linear functorial isomorphism commuting with φ and N

$$\alpha^i_{\mathrm{st}}(U) : H^i_{\text{ét}}(U, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \simeq H^i_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}},$$

which induces a B_{dR}-linear filtered isomorphism

$$H^i_{\text{ét}}(U, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{dR}} \simeq H^i_{B^+_{\mathrm{dR}}}(X) \otimes_{B^+_{\mathrm{dR}}} B_{\mathrm{dR}}.$$

*Here, $H^i_{B^+_{\mathrm{dR}}}(X)$ is the logarithmic B^+_{dR}-cohomology introduced in [**?**], where the log structure of X is induced by Z, and the filtration on $H^i_{B^+_{\mathrm{dR}}}(X)$ is defined by*

$$\mathrm{Fil}^\star H^i_{B^+_{\mathrm{dR}}}(X) := \mathrm{Im}(H^i(\mathrm{Fil}^\star \mathrm{R}\Gamma_{B^+_{\mathrm{dR}}}(X)) \to H^i_{B^+_{\mathrm{dR}}}(X)).$$

(2) Let $i \leq r$. Then we have an exact sequence

$$0 \to H^i_{\text{ét}}(U, \mathbb{Q}_p(r)) \to (H^i_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B^+_{\mathrm{st}})^{N=0, \varphi=p^r} \to H^i_{B^+_{\mathrm{dR}}}(X)/\mathrm{Fil}^r \to 0. \tag{1.1}$$

Moreover, when X descends to a rigid analytic variety over K, the statements in (1) and (2) are Galois-equivariant.

Remark 1.3. (1) In contrast to the general pro-étale setting, $H^i_{\text{ét}}(U, \mathbb{Q}_p)$ and $H^i_{\mathrm{HK}}(U)$ are finite-dimensional vector spaces.

(2) We have a natural isomorphism $H^i_{B^+_{\mathrm{dR}}}(X) \simeq H^i_{\mathrm{dR}}(U/B^+_{\mathrm{dR}})$, where $H^i_{\mathrm{dR}}(U/B^+_{\mathrm{dR}})$ is the B^+_{dR}-cohomology introduced in [**?**]. But the two cohomology groups have different filtrations.

(3) When X descends to a rigid analytic variety X_0 over K, we have a natural filtered isomorphism

$$H^i_{B^+_{\mathrm{dR}}}(X) = H^i_{\mathrm{logdR}}(X_0) \otimes_K B^+_{\mathrm{dR}}.$$

(4) The short exact sequence (**??**) is used in [**?**] to study the image of the Hodge-Tate log map.

(5) In a follow-up article [**?**], we will prove an étale version of the semistable conjecture for quasi-compact log rigid analytic varieties by introducing logarithmic syntomic cohomology. In particular, after extending the cohomology groups to the category of Vector Space, we can give another (more conceptual) proof of the above theorem.

We will discuss the proof of the semistable conjecture for étale cohomology in Section **??**, after introducing compactly supported Hyodo-Kato cohomology.

1.2. **Algebraic and analytic Hyodo-Kato cohomology.** Another foundational question is whether algebraic and analytic Hyodo-Kato cohomology can be compared, i.e., whether GAGA holds for Hyodo-Kato cohomology.

The main difficulty is that algebraic and analytic Hyodo-Kato cohomology are constructed differently. Briefly, the approach to defining Hyodo-Kato cohomology for rigid analytic varieties in [**?**] is similar to the algebraic construction in [**?**], by using the fact that smooth rigid analytic varieties admit semistable formal models étale locally [**?**]. However, unlike the algebraic case, in general there is no "compactification" for rigid analytic varieties, and we need to define Hyodo-Kato cohomology (locally) without adding horizontal divisors. This is the main problem when comparing the analytic and algebraic Hyodo-Kato cohomology, even if we have the Hyodo-Kato isomorphism in the geometric case.

To illustrate the difference between the algebraic and analytic definitions, consider the Hyodo-Kato cohomology for $\mathbb{A}^1_K$ and $\mathbb{A}^{1,\mathrm{an}}_K$, where $K=\mathbb{Q}_p$. For the algebraic affine line, let $(\mathbb{P}^1_{\mathcal{O}_K},\mathcal{M})$ be the log scheme $\mathbb{P}^1_{\mathcal{O}_K}$ with the log structure induced by the closed immersion $\{\infty\}\cup\mathbb{P}^1_k\hookrightarrow\mathbb{P}^1_{\mathcal{O}_K}$. Denote by $(\mathbb{P}^1_k,\mathcal{M}')$ the log scheme over $\mathrm{Spec}(k)$ induced by the base change $\mathrm{Spec}(k)\hookrightarrow\mathrm{Spec}(\mathcal{O}_K)$. Then the algebraic Hyodo-Kato cohomology of $\mathbb{A}^1_K$ can be computed by

$$\mathrm{R}\Gamma_{\mathrm{HK}}(\mathbb{A}^1_K)=\mathrm{R}\Gamma_{\mathrm{cris}}((\mathbb{P}^1_k,\mathcal{M}')^0/\mathcal{O}^0_K)_{\mathbb{Q}_p},$$

where $\mathcal{O}^0_K$ is the log scheme $\mathcal{O}_K$ with the log structure induced by $\mathbb{N}\to\mathcal{O}_K, 1\mapsto 0$.

On the other hand, to compute $\mathrm{R}\Gamma_{\mathrm{HK}}(\mathbb{A}^{1,\mathrm{an}}_K)$, by definition one has to cover it by the rigid generic fibers of semistable formal models and then use descent. For example, let $\mathfrak{X}$ be the standard semistable formal model of $\mathbb{A}^{1,\mathrm{an}}_K$, i.e., the one coming from the Bruhat-Tits building of the rigid analytic projective line. Let $\mathfrak{X}_n\subset\mathfrak{X}$ be the induced semistable formal model of $X_n:=\mathrm{Sp}(K\langle\omega^n x\rangle)$, the closed disk of radius $|\omega|^{-n}$, where ω is a uniformizer of K. Then the Hyodo-Kato cohomology $\mathrm{R}\Gamma_{\mathrm{HK}}(\mathbb{A}^{1,\mathrm{an}}_K)$ is computed by

$$\mathrm{R}\Gamma_{\mathrm{HK}}(\mathbb{A}^{1,\mathrm{an}}_K)=R\lim_n\mathrm{R}\Gamma_{\mathrm{cris}}(\mathfrak{X}^0_{n,k}/\mathcal{O}^0_K)_{\mathbb{Q}_p}.$$

There is no evident direct construction of a morphism $\mathrm{R}\Gamma_{\mathrm{HK}}(\mathbb{A}^1_K)\to\mathrm{R}\Gamma_{\mathrm{HK}}(\mathbb{A}^{1,\mathrm{an}}_K)$, although one can show that both sides have vanishing cohomology in nonzero degrees in this case.

In this article, by studying the geometric properties of algebraic and analytic Hyodo-Kato cohomology, we will show that we indeed have GAGA for Hyodo-Kato cohomology:

Theorem 1.4. *(Theorem **??**) (1) Let X be an algebraic variety over K, and let X^{an} be its analytification. Then there exists a natural quasi-isomorphism in $D(\mathrm{Mod}^{\mathrm{solid}}_F)$:*

$$\mathrm{R}\Gamma_{\mathrm{HK}}(X)\xrightarrow{\simeq}\mathrm{R}\Gamma_{\mathrm{HK}}(X^{\mathrm{an}}),$$

which is compatible with Frobenius, monodromy, and the GAGA morphism for de Rham cohomology, i.e., we have the following commutative square:

$$\begin{array}{ccc}\mathrm{R}\Gamma_{\mathrm{HK}}(X) & \xrightarrow{\iota_{\mathrm{HK}}} & \mathrm{R}\Gamma_{\mathrm{dR}}(X)\\ \downarrow\simeq & & \downarrow\simeq\\ \mathrm{R}\Gamma_{\mathrm{HK}}(X^{\mathrm{an}}) & \xrightarrow{\iota_{\mathrm{HK}}} & \mathrm{R}\Gamma_{\mathrm{dR}}(X^{\mathrm{an}}),\end{array}$$

where the horizontal maps are Hyodo-Kato morphisms.

(2) Let X be an algebraic variety over $\overline{K}$, and let X^{an}_C be the analytification of $X_C:=X\times_{\overline{K}}C$. Then there exists a natural quasi-isomorphism in $D(\mathrm{Mod}^{\mathrm{solid}}_{F^{\mathrm{nr}}})$:

$$\mathrm{R}\Gamma_{\mathrm{HK}}(X)\xrightarrow{\simeq}\mathrm{R}\Gamma_{\mathrm{HK}}(X^{\mathrm{an}}_C),$$

which is compatible with the Galois action (when X descends to K), Frobenius, monodromy, and the GAGA morphism for de Rham cohomology.

Remark 1.5. This theorem is also used in [**?**] to study the image of the Hodge-Tate log map.

1.3. **Cohomology with compact support.** To construct the comparison map between algebraic and analytic Hyodo-Kato cohomology, we first note that if $\mathcal{X}$ is a projective semistable scheme over $\mathcal{O}_K$, then local-global compatibility gives natural identifications

$$\mathrm{R}\Gamma_{\mathrm{HK}}(\mathcal{X}_K) = \mathrm{R}\Gamma_{\mathrm{HK}}(\mathcal{X}_K^{\mathrm{an}}) \simeq \mathrm{R}\Gamma_{\mathrm{cris}}(\mathcal{X}_k^0/\mathcal{O}_K^0)_{\mathbb{Q}_p}.$$

We would like to reduce the proof of GAGA to the projective case (with a semistable model). As in the example in the preceding subsection, in general one needs to work on a strictly semistable pair. Briefly, a semistable pair $(U, \overline{U})$ over a field K is an open embedding $j : U \hookrightarrow \overline{U}$ with dense image of a K-variety U into a proper flat regular $\mathcal{O}_K$-scheme $\overline{U}$, such that $\overline{U} - U$ is a divisor with normal crossings on $\overline{U}$, the irreducible components of $\overline{U} - U$ are regular, and the special fiber $\overline{U}_k$ is reduced. Let $D = \overline{U}_K - U$. If we have the compatible Gysin sequences

$$\mathrm{R}\Gamma_{\mathrm{HK}}(D)\{-1\}[-2] \simeq [\mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_K) \to \mathrm{R}\Gamma_{\mathrm{HK}}(U)],$$

$$\mathrm{R}\Gamma_{\mathrm{HK}}(D^{\mathrm{an}})\{-1\}[-2] \simeq [\mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_K^{\mathrm{an}}) \to \mathrm{R}\Gamma_{\mathrm{HK}}(U^{\mathrm{an}})]$$

where D is smooth, then since $\overline{U}_K$ and D are proper, we should be able to deduce the GAGA comparison for U and then pass to the general case.

The algebraic Gysin sequence follows from the realization functor for (geometric) Hyodo-Kato cohomology constructed in [**?**]. However, it is still unclear how to construct the Gysin morphism directly in rigid geometry. On the other hand, the Gysin morphism can be deduced from Poincaré duality and the open-closed exact sequence of compactly supported Hyodo-Kato cohomology. We will review the definition of compactly supported Hyodo-Kato cohomology, which is developed in [**?**].

Therefore, the proof of GAGA heavily relies on the following propositions.

Proposition 1.6 (open-closed exact sequence)**.** *(Proposition* **??***) Let X be a smooth dagger variety over $L = K$ or C, let $Z \subset X$ be a smooth divisor, and set $U = X - Z$. Then we have*

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(U) \simeq [\mathrm{R}\Gamma_{\mathrm{HK,c}}(X) \to \mathrm{R}\Gamma_{\mathrm{HK,c}}(Z)].$$

Theorem 1.7. *(*[**?**, Theorems 5.30 and 5.34]*; see also Theorem* **??***) Let Y be a partially proper smooth rigid analytic variety or a quasi-compact smooth dagger variety over C of dimension d. Then:*

(i) There is a natural trace map in $D(\mathrm{Mod}^{\mathrm{solid}}_{F^{\mathrm{nr}}})$:

$$\mathrm{tr}_{\mathrm{HK}} : \mathrm{R}\Gamma_{\mathrm{HK,c}}(Y)[2d] \to F^{\mathrm{nr}}\{-d\},$$

compatible with the Hyodo-Kato morphism.

(ii) The pairing

$$\mathrm{R}\Gamma_{\mathrm{HK}}(Y) \otimes^{L_{\blacksquare}} \mathrm{R}\Gamma_{\mathrm{HK,c}}(Y)[2d] \to \mathrm{R}\Gamma_{\mathrm{HK,c}}(Y)[2d] \to F^{\mathrm{nr}}\{-d\}$$

is a perfect duality in $D(\mathrm{Mod}^{\mathrm{solid}}_{F^{\mathrm{nr}}})$, i.e., we have the induced quasi-isomorphism in $D(\mathrm{Mod}^{\mathrm{solid}}_{F^{\mathrm{nr}}})$:

$$\mathrm{R}\Gamma_{\mathrm{HK}}(Y) \simeq R\,\mathrm{Hom}_{F^{\mathrm{nr}}}(\mathrm{R}\Gamma_{\mathrm{HK,c}}(Y), F^{\mathrm{nr}}\{-d\}[-2d]).$$

1.4. **The proof of the semistable conjecture.** As in the proof of GAGA, our strategy is to reduce the semistable conjecture for almost proper rigid analytic varieties to the proper case. The proof includes the following steps:

- Construction of the period morphism: in [**?**, Theorem 6.9], for $r \geq 0$, Pierre Colmez and Wiesława Nizioł construct a natural quasi-isomorphism
$$\tau^{\leq r} \mathrm{R}\Gamma_{\mathrm{syn}}(U, \mathbb{Q}_p(r)) \xrightarrow{\simeq} \tau^{\leq r} \mathrm{R}\Gamma_{\mathrm{pro\acute{e}t}}(U, \mathbb{Q}_p(r))$$
for any partially proper rigid analytic variety, and in our case the syntomic cohomology $\mathrm{R}\Gamma_{\mathrm{syn}}(U, \mathbb{Q}_p(r))$ can be described as
$$\mathrm{R}\Gamma_{\mathrm{syn}}(U, \mathbb{Q}_p(r)) := [[\mathrm{R}\Gamma_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}^+]^{N=0, \varphi=p^r} \to \mathrm{R}\Gamma_{\mathrm{dR}}(U/B_{\mathrm{dR}}^+)/\operatorname{Fil}^r], \tag{1.2}$$
For $r > 2d$, where $d = \dim U$, this induces a map, using the inclusion $B_{\mathrm{st}}^+ \subset B_{\mathrm{st}}$:
$$\tau^{\leq 2d} \mathrm{R}\Gamma_{\mathrm{pro\acute{e}t}}(U, \mathbb{Q}_p(r)) \xrightarrow{\simeq} \tau^{\leq 2d} \mathrm{R}\Gamma_{\mathrm{syn}}(U, \mathbb{Q}_p(r)) \to [\mathrm{R}\Gamma_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}]^{N=0, \varphi=p^r} \xrightarrow{p^{-r}} \mathrm{R}\Gamma_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}.$$
After imposing B_{st}-linearity and twisting, this allows us to define the period morphism
$$\alpha_{\mathrm{st}}(U) : \mathrm{R}\Gamma_{\text{ét}}(U, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \to \mathrm{R}\Gamma_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}},$$
by simply applying the natural morphism $\mathrm{R}\Gamma_{\text{ét}}(U, \mathbb{Q}_p) \to \mathrm{R}\Gamma_{\mathrm{pro\acute{e}t}}(U, \mathbb{Q}_p)$. This construction might seem strange, as $\mathrm{R}\Gamma_{\text{ét}}(U, \mathbb{Q}_p)$ and $\mathrm{R}\Gamma_{\mathrm{pro\acute{e}t}}(U, \mathbb{Q}_p)$ are generally quite different. For example, when $U = \mathbb{A}_C^{1,\mathrm{an}}$, the analytic affine line, $H^1_{\text{ét}}(U, \mathbb{Q}_p) = 0$ but $H^1_{\mathrm{pro\acute{e}t}}(U, \mathbb{Q}_p)$ is very large, as indicated by the main theorem of [**?**]. Nevertheless, this construction yields precisely the period morphism required. The same method also defines $\alpha_{\mathrm{st}}(X)$: when X is proper and smooth, $\mathrm{R}\Gamma_{\text{ét}}(X, \mathbb{Q}_p) = \mathrm{R}\Gamma_{\mathrm{pro\acute{e}t}}(X, \mathbb{Q}_p)$, and Pierre Colmez and Wiesława Nizioł prove that $\alpha_{\mathrm{st}}(X)$ is a quasi-isomorphism in [**?**], by using the theory of Banach-Colmez spaces.
- Reducing to the proper case: we can reduce to the case where Z is smooth, and we check that α_{st} is compatible with the Gysin sequence. Then we can apply the Gysin sequence to show that $\alpha_{\mathrm{st}}(U)$ is a natural quasi-isomorphism. The key point is that our construction of α_{st} is natural with respect to $U \to X$, so it suffices to check that α_{st} is compatible with the Gysin morphism, i.e., that the following square
$$\begin{array}{ccc} \mathrm{R}\Gamma_{\text{ét}}(Z, \mathbb{Q}_p(-1))[-2] \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} & \xrightarrow{g_{\text{ét}}} & \mathrm{R}\Gamma_{\text{ét}}(X, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \\ \downarrow{\scriptstyle \alpha_{\mathrm{st}}(Z)(-1)[-2]} & & \downarrow{\scriptstyle \alpha_{\mathrm{st}}(X)} \\ \mathrm{R}\Gamma_{\mathrm{HK}}(Z)\{-1\}[-2] \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}} & \xrightarrow{g_{\mathrm{HK}}} & \mathrm{R}\Gamma_{\mathrm{HK}}(X) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}, \end{array}$$
is commutative, where the horizontal maps are the Gysin morphisms. The Gysin morphisms are deduced from Poincaré duality and the open-closed exact sequence, and we can reduce to checking that α_{st} is compatible with the Poincaré duality, that is to say, checking that α_{st} is compatible with the pairing and the trace map, but both are automatic due to our construction of Poincaré duality for pro-étale cohomology (and hence for étale cohomology, since X and Z are proper smooth): in fact, the pairing and the trace map for pro-étale

cohomology are induced from those for Hyodo-Kato cohomology, by using the distinguished triangle (**??**).

- Construction of the short exact sequence (**??**): we construct (**??**) by using [**?**, Remark 5.16], which is an extension of the standard result for admissible filtered (φ, N)-modules: once we are able to show that $\alpha_{\mathrm{st}}(U)$ is a natural quasi-isomorphism, we can show that, for $i \leq r$,
$$(D, D^+_{\mathrm{dR}}) := (H^i_{\mathrm{HK}}(U), H^i_{\text{ét}}(U, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B^+_{\mathrm{dR}})$$
is a weakly admissible filtered (φ, N)-module over C (in the sense of [**?**, Definition 5.3.1]) with φ-slopes in $[0, r]$. The key observation here is that we have
$$(D \otimes_{F^{\mathrm{nr}}} B^+_{\mathrm{dR}})/t^r D^+_{\mathrm{dR}} \simeq H^i_{\mathrm{dR}}(U/B^+_{\mathrm{dR}})/t^r D^+_{\mathrm{dR}} \simeq H^i_{B^+_{\mathrm{dR}}}(X)/\operatorname{Fil}^r,$$
where the first isomorphism is the geometric Hyodo-Kato isomorphism [**?**, Theorem 4.6], and the second isomorphism follows from [**?**, Proposition 5.17] and [**?**, Theorem 1.4]. Then [**?**, Remark 5.16] gives the desired short exact sequence (**??**).
- Checking that our construction of (**??**) is Galois-equivariant: however, the previous step intertwines different constructions of period morphisms, which could affect compatibility with the Galois action when X descends to a rigid analytic variety over K. To resolve this issue, we need to show that our period map $\alpha_{\mathrm{st}}(U)$, after tensoring with B_{dR}, agrees with the de Rham period map of [**?**, Theorem 1.4]. Once again, we can reduce to Z smooth, and by using the Gysin sequence, we are reduced to checking that the maps $\alpha_{\mathrm{st}}(X) \otimes_{B_{\mathrm{st}}} B_{\mathrm{dR}}$ and $\alpha_{\mathrm{st}}(Z) \otimes_{B_{\mathrm{st}}} B_{\mathrm{dR}}$ agree with the de Rham period morphism for proper smooth rigid analytic varieties constructed by Peter Scholze in [**?**], but this compatibility has been established by Sally Gilles in [**?**].

We refer the reader to Theorem **??** for the detailed proof.

1.5. **Structure of the article.** In Section 2, we review the h-topology from [**?**] and the éh-topology from [**?**], along with the (log-)de Rham comparison theorem. These tools are fundamental for defining Hyodo-Kato cohomology and proving the semistable conjecture.

In Section 3, we present some key aspects of both algebraic and analytic Hyodo-Kato cohomology, including their compactly supported versions. We define the algebraic Hyodo-Kato cohomology with compact support, which agrees with the construction via the Hyodo-Kato realization functor from [**?**]. For the analytic case, we review the compactly supported analytic Hyodo-Kato cohomology as defined in [**?**].

Section 4 focuses on the geometric properties of compactly supported Hyodo-Kato cohomology for analytic varieties. We establish the Mayer-Vietoris properties for Hyodo-Kato cohomology (and related theories) in the context of abstract blow-up squares. In the absence of a six-functor formalism within the rigid analytic framework (which is still under development), we demonstrate Poincaré duality and the open-closed exact sequence for compactly supported Hyodo-Kato cohomology as defined here. This will also yield the Gysin sequence.

In Section 5, we apply the geometric properties developed in Section 4 to establish key results in Hyodo-Kato cohomology. As outlined in the introduction, we compare algebraic and analytic Hyodo-Kato cohomology and establish the semistable conjecture for almost proper rigid analytic varieties. We will also examine the construction of log-crystalline cohomology with compact support

by Tsuji in [**?**], showing that our definition of compactly supported Hyodo-Kato cohomology agrees with Tsuji's construction.

Finally, in the appendix, we verify that the overconvergent de Rham cohomology defined for possibly singular dagger varieties in [**?**] agrees with the definition in [**?**]. This result is used in Section 4 and enables us to confirm the finiteness of overconvergent Hyodo-Kato cohomology in broader cases.

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Notation and conventions. In this article, we use the language of adic spaces developed in [**?**]. A rigid analytic variety is a quasi-separated adic space locally of finite type over $\mathrm{Spa}(L, \mathcal{O}_L)$ for a p-adic field L. All rigid analytic spaces considered will be over K or C. We will also use the notion of dagger varieties [**?**]. We assume all rigid analytic varieties and dagger varieties are separated, taut, and countable at infinity. We denote by Rig_L (or $\mathrm{Rig}_L^\dagger$) the category of rigid analytic varieties (or dagger varieties) over L, and we denote by Sm_L (or $\mathrm{Sm}_L^\dagger$) the category of smooth rigid analytic varieties (or smooth dagger varieties) over L.

If $\mathcal{A}$ is an abelian category, unless otherwise stated, we always work with the derived stable ∞-category $D(\mathcal{A})$.

We will freely use the language of condensed mathematics developed in [**?**]. In fact, most of the cohomology groups appearing in this article are finite-dimensional vector spaces, so this will not cause much trouble.

We will also use the theory of log adic spaces. For definitions and properties of log adic spaces, see [**?**]. A (pre)log-structure on a condensed ring is simply a (pre)log-structure on the underlying ring.

2. Preliminaries

In this section, we review the preliminaries needed to define Hyodo-Kato cohomology and prove the semistable conjecture for étale cohomology of almost proper rigid analytic varieties.

2.1. The éh-topology and h-topology.

The éh-topology is used to define Hyodo-Kato cohomology for singular rigid analytic varieties. It admits a basis of semistable formal models by η-étale descent. On the algebraic side, there is a similar definition of the éh-topology, but to obtain a basis of semistable models, we need to refine it to the h-topology. We recall the definitions of the éh-topology for rigid analytic varieties and the h-topology for algebraic varieties. We begin with a generalization of Verdier's criterion.

2.1.1. *Beilinson basis.* In [**?**], Beilinson extended Verdier's well-known criterion [**?**, 4.1], which provides conditions for changing sites while preserving their associated topoi. We briefly review this result.

Let $\mathcal{C}$ be an essentially small site. We denote by $\mathrm{Sh}(\mathcal{C})$ the category of sheaves of sets on $\mathcal{C}$. A (Beilinson) basis is defined as follows.

Definition 2.1. A basis of $\mathcal{C}$ is a pair $(\mathcal{B}, \phi)$, where $\mathcal{B}$ is an essentially small category and $\phi : \mathcal{B} \to \mathcal{C}$ is a faithful functor with the following property: for every $C \in \mathcal{C}$ and every finite family of pairs (B_α, f_α), with $B_\alpha \in \mathcal{B}$ and $f_\alpha : C \to \phi(B_\alpha)$ for $\alpha \in I$, there exist objects $B'_\beta \in \mathcal{B}$ and a covering family $\{\phi(B'_\beta) \to C\}$ such that each composition $\phi(B'_\beta) \to C \to \phi(B_\alpha)$ lies in $\mathrm{Hom}_{\mathcal{B}}(B'_\beta, B_\alpha) \subset \mathrm{Hom}_{\mathcal{C}}(\phi(B'_\beta), \phi(B_\alpha))$.

Remark 2.2. Verdier's criterion requires ϕ to be fully faithful, whereas here we only require it to be faithful. This is useful when considering a basis of semistable models, where ϕ is the (rigid) generic fiber functor.

A sieve in $\mathcal{B}$ is called covering if its image under ϕ generates a covering sieve in $\mathcal{C}$. We have the following theorem from [**?**].

Theorem 2.3. *If $(\mathcal{B}, \phi)$ is a basis of $\mathcal{C}$, then*
(i) Covering sieves in $\mathcal{B}$ define a Grothendieck topology on $\mathcal{B}$.
(ii) The functor $\phi : \mathcal{B} \to \mathcal{C}$ is continuous.
(iii) ϕ induces an equivalence of topoi $\mathrm{Sh}(\mathcal{B}) \xrightarrow{\simeq} \mathrm{Sh}(\mathcal{C})$.

2.1.2. *h-topology for algebraic varieties.* We begin with the definition of the h-topology from [**?**].

Definition 2.4. For a field L, the h-topology on Var_L is generated by the pretopology whose coverings are finite families of maps $\{Y_i \to X\}_{i \in I}$ such that $Y := \coprod_{i \in I} Y_i \to X$ is a universal topological epimorphism, i.e., a subset U of X is Zariski open if and only if its preimage in Y is open, and the same is true after any base change over X.

Remark 2.5. The h-topology is finer than the proper and étale topologies, but coarser than the v-topology.

A pair $(U, \overline{U})$ over K consists of a K-variety U, a reduced proper flat $\mathcal{O}_K$-scheme $\overline{U}$, and an open embedding $j : U \hookrightarrow \overline{U}$ with dense image. We denote by $\mathrm{Var}_K^{\mathrm{ss}}$ the category of strictly semistable pairs, i.e., pairs $(U, \overline{U})$ such that $\overline{U}$ is regular, $\overline{U} - U$ is a divisor with normal crossings on $\overline{U}$, its irreducible components are regular, and the special fiber $\overline{U}_k$ is reduced.

In the geometric setting, we denote by $\mathrm{Var}_{\overline{K}}^{\mathrm{ss},b}$ the category of basic semistable $\mathcal{O}_{\overline{K}}$-pairs, i.e., semistable $\mathcal{O}_{\overline{K}}$-pairs $(U, \overline{U})$ for which there exist a finite extension E of K and a semistable pair $(U', \overline{U}')$ over $\mathcal{O}_E$ such that $(U, \overline{U})$ is isomorphic to the base change $(U'_{\overline{K}}, \overline{U}'_{\mathcal{O}_{\overline{K}}})$.

By [**?**, Proposition 2.5], we have

Proposition 2.6. *(i) $\mathrm{Var}_K^{\mathrm{ss}}$ forms a basis of $\mathrm{Var}_{K,\mathrm{h}}$.*
(ii) Each of $\mathrm{Var}_{\overline{K}}^{\mathrm{ss},b}$ and $\mathrm{Var}_{\overline{K}}^{\mathrm{ss}}$ forms a basis of $\mathrm{Var}_{\overline{K},\mathrm{h}}$.

We will often use the following observation.

Lemma 2.7. *Suppose $(U, \overline{U})$ is a strictly semistable pair, and let $Z := \overline{U}_\eta - U$. If Z is smooth and $\overline{Z}$ denotes its closure in $\overline{U}$, then $(Z, \overline{Z})$ also belongs to $\mathrm{Var}_K^{\mathrm{ss}}$.*

Proof. Since $(U, \overline{U})$ is a strictly semistable pair, we can write $\overline{U} - U = \overline{Z} \cup D_1 \cup \cdots \cup D_m$, where the D_i are the irreducible components contained in the special fiber. For $D_J = \bigcap_{j\in J} D_j$, where $J \subset \{1, 2, \ldots, m\}$, the intersection $\overline{Z} \cap D_J$ is a regular scheme each of whose irreducible components has codimension $|J|$ in $\overline{Z}$. Moreover, $\overline{Z} - Z = \bigcup_i (\overline{Z} \cap D_i)$, and each $\overline{Z} \cap D_i$ is smooth over k. Write $\overline{Z} \cap D_i$ as a disjoint union of smooth irreducible components $\coprod_k D_{ik}$. For a finite set J' of indices, the intersection $D'_{J'} = \bigcap_{(j,l)\in J'} D_{jl}$ is regular, and each of its irreducible components has codimension $|J'|$ in $\overline{Z}$. Thus $\overline{Z} - Z$ is a divisor with strict normal crossings in $\overline{Z}$. □

2.1.3. *éh-topology for rigid analytic varieties.* We briefly summarize [**?**, Section 2].

Definition 2.8. For a field L, the éh-topology on Rig_L is generated by the pretopology whose coverings can be refined by étale coverings, universal homeomorphisms, and morphisms

$$\mathrm{Bl}_Z(X) \sqcup Z \to X,$$

where Z is a closed analytic subset of X.

We denote by $\mathscr{M}_K^{\mathrm{ss}}$ the category of formal $\mathcal{O}_K$-models that are semistable over $\mathcal{O}_E$ for some finite extension E of K. We also denote by $\mathscr{M}_C^{\mathrm{ss},b}$ the category of basic semistable formal $\mathcal{O}_C$-models, i.e., formal $\mathcal{O}_C$-models $\mathscr{X}$ for which there exist a finite extension E of K and a semistable model $\mathscr{X}'$ over $\mathcal{O}_E$ such that $\mathscr{X}$ is isomorphic to the base change $\mathscr{X}'_{\mathcal{O}_C}$. By [**?**, Proposition 2.8] and resolution of singularities, we have the following proposition.

Proposition 2.9. *(i) Each of* Sm_K *and* $\mathscr{M}_K^{\mathrm{ss}}$ *forms a basis of* $\mathrm{Rig}_{K,\text{éh}}$.
(ii) Over C*, each of* Sm_C*,* $\mathscr{M}_C^{\mathrm{ss},b}$*, and* $\mathscr{M}_K^{\mathrm{ss}}$ *forms a basis of* $\mathrm{Rig}_{C,\text{éh}}$.

There are analogous results for dagger varieties. We denote by $\mathscr{M}_L^{\mathrm{wss}}$ the category of weakly semistable formal $\mathcal{O}_L$-models, and by $\mathscr{M}_C^{\mathrm{wss},b}$ the category of basic weakly semistable formal $\mathcal{O}_C$-models. By [**?**, Proposition 2.14], we have the following proposition.

Proposition 2.10. *(i) Each of* $\mathrm{Sm}_K^\dagger$ *and* $\mathscr{M}_K^{\mathrm{wss}}$ *forms a basis of* $\mathrm{Rig}_{K,\text{éh}}^\dagger$.
(ii) Over C*, each of* $\mathrm{Sm}_C^\dagger$*,* $\mathscr{M}_C^{\mathrm{wss},b}$*, and* $\mathscr{M}_K^{\mathrm{wss}}$ *forms a basis of* $\mathrm{Rig}_{C,\text{éh}}^\dagger$.

2.2. **de Rham cohomology and B_{dR}^+-cohomology.** For $i \geq 0$, let $\Omega^i_{\text{éh}}$ be the éh-sheafification of the presheaf $X \mapsto \Omega^i(X)$ for $X \in \mathrm{Sm}_L$ or $\mathrm{Sm}_L^\dagger$. We have the following éh-descent theorem from [**?**, Theorem 4.0.2].

Theorem 2.11. *If* $X \in \mathrm{Sm}_L$ *or* $\mathrm{Sm}_L^\dagger$*, then for any* $i \in \mathbb{N}$*, we have*

$$R\pi_{X,*}\Omega^i_{\text{éh}} = \Omega^i_{X/L}.$$

Using the éh-topology, we define de Rham cohomology for rigid analytic varieties as follows.

Definition 2.12. Let X be a rigid analytic variety or a dagger variety over L. Its de Rham cohomology is defined by

$$\mathrm{R}\Gamma_{\mathrm{dR}}(X) := \mathrm{R}\Gamma(X, \Omega^\bullet_{\text{éh}}).$$

For X defined over C, a meaningful de Rham-étale comparison theorem requires B^+_{dR}-cohomology, which is a deformation of de Rham cohomology:

$$H^i_{\mathrm{dR}}(X/B^+_{\mathrm{dR}}) \otimes_{B^+_{\mathrm{dR}}} C \simeq H^i_{\mathrm{dR}}(X).$$

We refer the reader to [**?**], [**?**], [**?**], and [**?**] for various constructions of B^+_{dR}-cohomology.

The following theorem is due to [**?**, Theorem 1.1.4] and [**?**, Theorem 7.4].

Theorem 2.13. *Let X be a proper rigid analytic variety. For each $i \geq 0$, the following statements hold.*

(i) If X is defined over K, we have a natural isomorphism

$$H^i_{\text{ét}}(X_C, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{dR}} \simeq H^i_{\mathrm{dR}}(X) \otimes_K B_{\mathrm{dR}},$$

compatible with filtrations.

(ii) If X is defined over C, we have a natural isomorphism

$$H^i_{\text{ét}}(X, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{dR}} \simeq H^i_{\mathrm{dR}}(X/B^+_{\mathrm{dR}}) \otimes_{B^+_{\mathrm{dR}}} B_{\mathrm{dR}},$$

compatible with filtrations. When X descends to X_0 over K, this isomorphism agrees with the comparison isomorphism in (i), under the canonical identification

$$H^i_{\mathrm{dR}}(X/B^+_{\mathrm{dR}}) = H^i_{\mathrm{dR}}(X_0) \otimes_K B^+_{\mathrm{dR}}.$$

The following logarithmic analogue for proper smooth rigid analytic varieties generalizes [**?**, Theorem 1.7].

Theorem 2.14. [**?**] *Let X be a proper smooth rigid analytic variety over C, let $D \subset X$ be a divisor with strict normal crossings, and set $U = X - D$. We endow X with the logarithmic structure induced by D. Then there are cohomology groups $H^i_{B^+_{\mathrm{dR}}}(X)$ equipped with a canonical filtered isomorphism*

$$H^i_{\text{két}}(X, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{dR}} \simeq H^i_{B^+_{\mathrm{dR}}}(X) \otimes_{B^+_{\mathrm{dR}}} B_{\mathrm{dR}}.$$

When (X, D) descends to (X_0, D_0) over a discretely valued field K, this isomorphism agrees with the comparison isomorphism (see [**?**, Theorem 1.1]*)*

$$H^i_{\text{két}}(X, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{dR}} \simeq H^i_{\mathrm{logdR}}(X_0) \otimes_K B_{\mathrm{dR}},$$

under the canonical identification

$$H^i_{B^+_{\mathrm{dR}}}(X) = H^i_{\mathrm{logdR}}(X_0) \otimes_K B^+_{\mathrm{dR}}.$$

Moreover, $H^i_{B^+_{\mathrm{dR}}}(X)$ is a finite free B^+_{dR}-module, and the following statements hold.

(i) The Hodge–de Rham spectral sequence

$$E_1^{ij} := H^j(X_{\log}, \Omega_X^{\log,i}) \Rightarrow H^{i+j}_{\mathrm{logdR}}(X)$$

degenerates at E_1.

(ii) The Hodge–Tate spectral sequence

$$E_2^{ij} := H^j(X_{\log}, \Omega_X^{\log,i})(-i) \Rightarrow H^{i+j}_{\text{két}}(X, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} C$$

degenerates at E_2.

2.3. **Cohomology for dagger varieties.** Following [**?**] and [**?**], we briefly review a construction that allows us to canonically define a cohomology theory on $\mathrm{Rig}_L^\dagger$ from a cohomology theory on Rig_L.

Let $\mathcal{D}$ be a presentable ∞-category. The continuous functor

$$l : \mathrm{Rig}^\dagger_{L,\text{éh}} \to \mathrm{Rig}_{L,\text{éh}} : X \mapsto \widehat{X}$$

given by sending a dagger variety X to its completion $\widehat{X}$ induces an adjunction

$$l_* : \mathrm{Shv}^{\mathrm{hyp}}(\mathrm{Rig}^\dagger_{L,\text{éh}}, \mathcal{D}) \rightleftarrows \mathrm{Shv}^{\mathrm{hyp}}(\mathrm{Rig}_{L,\text{éh}}, \mathcal{D}) : l^{*\mathrm{hyp}},$$

where $l^{*\mathrm{hyp}}$ is the hypercompletion of l^*. For $\mathcal{F} \in \mathrm{Shv}^{\mathrm{hyp}}(\mathrm{Rig}_{L,\text{éh}}, \mathcal{D})$, denote by

$$\mathcal{F}^\dagger := l^{*\mathrm{hyp}}\mathcal{F} \in \mathrm{Shv}^{\mathrm{hyp}}(\mathrm{Rig}^\dagger_{L,\text{éh}}, \mathcal{D}).$$

In what follows, whenever we introduce a cohomology theory without compact support for rigid analytic varieties (e.g., de Rham, Hyodo-Kato, étale, pro-étale, or syntomic cohomology), we apply this construction to define the corresponding theory for dagger varieties, taking $\mathcal{D} = D(\mathrm{Mod}^{\mathrm{solid}}_{\mathbb{Q}})$.

The étale cohomology of a dagger variety X agrees with that of $\widehat{X}$, by [**?**, Corollary A.28].

3. Hyodo-Kato cohomology (with compact support)

In this section we review the definitions of Hyodo-Kato cohomology for algebraic varieties in [**?**] and for rigid analytic varieties in [**?**]. We also define Hyodo-Kato cohomology with compact support. In the following sections, we will show that our definition agrees with Tsuji's log-crystalline cohomology with compact support in [**?**].

3.1. **Hyodo-Kato cohomology for algebraic varieties.** We review the Hyodo-Kato cohomology (with compact support) for algebraic varieties. In particular, thanks to the Hyodo-Kato realization functor, we have Poincaré duality for Hyodo-Kato cohomology.

3.1.1. *Definition and properties.* We begin by recalling the definition and basic properties of algebraic Hyodo-Kato cohomology from [**?**] and [**?**].

(1) Arithmetic setup: Here we define Hyodo-Kato cohomology of algebraic varieties over K. Suppose $(U, \overline{U}) \in \mathrm{Var}_K^{\mathrm{ss}}$ is a strictly semistable pair, as described before Proposition **??**. Let $\mathscr{A}_{\mathrm{HK}}$ be the h-sheafification of the presheaf

$$(U, \overline{U}) \mapsto \mathrm{R}\Gamma_{\mathrm{HK}}((U, \overline{U})_0) := \mathrm{R}\Gamma_{\mathrm{cris}}((U, \overline{U})_0/\mathcal{O}^0_{F_L})_{\mathbb{Q}_p},$$

where we endow $(U, \overline{U})$ with the compactifying log structure associated with the open immersion $U \hookrightarrow \overline{U}$. The sheaf $\mathscr{A}_{\mathrm{HK}}$ is an h-sheaf of dg solid F-algebras on Var_K equipped with a Frobenius operator ϕ and a monodromy operator N such that $N\phi = p\phi N$. For an algebraic variety $X \in \mathrm{Var}_K$, we define the Hyodo-Kato cohomology of X by $\mathrm{R}\Gamma_{\mathrm{HK}}(X) := \mathrm{R}\Gamma(X, \mathscr{A}_{\mathrm{HK}})$. By [**?**], there exists a Hyodo-Kato morphism

$$\iota_{\mathrm{HK}} : \mathrm{R}\Gamma_{\mathrm{HK}}(X) \to \mathrm{R}\Gamma_{\mathrm{dR}}(X),$$

which is constructed locally for semistable pairs and then descended.

(2) Geometric setup: Suppose now that $(U, \overline{U}) \in \mathrm{Var}_{\overline{K}}^{\mathrm{b,ss}}$ is a basic semistable model. Endow $(U, \overline{U})$ with the log structure as above. Then, by definition, $f : (U, \overline{U}) \to \mathrm{Spec}(\mathcal{O}_{\overline{K}})^{\times}$ is the base change of a semistable pair $\theta_L : (U_L, \overline{U_{\mathcal{O}_L}}) \to \mathrm{Spec}(\mathcal{O}_L)^{\times}$ for a finite field extension L/K. The data $(L, \theta_L, U_L, \overline{U_{\mathcal{O}_L}})$ form a filtered set Σ. Note that for a morphism $(L, \theta_L, U_L, \overline{U_{\mathcal{O}_L}}) \to (L', \theta_{L'}, U'_{L'}, \overline{U'_{\mathcal{O}_{L'}}})$ in Σ, the base change theorem tells us

$$\mathrm{R}\Gamma_{\mathrm{HK}}((U'_{L'}, \overline{U'_{\mathcal{O}_{L'}}})_0) \otimes_{F'_L} F_L \simeq \mathrm{R}\Gamma_{\mathrm{HK}}((U_L, \overline{U_{\mathcal{O}_L}})_0).$$

Let $\mathscr{A}_{\mathrm{HK}}$ be the h-sheafification of the presheaf

$$(U, \overline{U}) \mapsto \mathrm{R}\Gamma_{\mathrm{HK}}((U, \overline{U})_0) := \varinjlim_{L} \mathrm{R}\Gamma_{\mathrm{HK}}((U_L, \overline{U_{\mathcal{O}_L}})_0),$$

where the filtered colimit is indexed by Σ. This is an h-sheaf of dg solid F^{nr}-algebras on $\mathrm{Var}_{\overline{K}}$ equipped with a Frobenius operator ϕ and a monodromy operator N such that $N\phi = p\phi N$. For an algebraic variety $X \in \mathrm{Var}_{\overline{K}}$, we define the Hyodo-Kato cohomology of X by $\mathrm{R}\Gamma_{\mathrm{HK}}(X) := \mathrm{R}\Gamma(X, \mathscr{A}_{\mathrm{HK}})$. By keeping more careful track of the data, Beilinson proved in [**?**] that, after fixing a pseudo-uniformizer, we have a natural Hyodo-Kato isomorphism:

$$\iota_{\mathrm{HK}} \otimes_{F^{\mathrm{nr}}} \overline{K} : \mathrm{R}\Gamma_{\mathrm{HK}}(X) \otimes^{L}_{F^{\mathrm{nr}}} \overline{K} \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{dR}}(X).$$

We have the following local-global compatibility.

Proposition 3.1. *(i)* [**?**, 2.5] *For any $(V, \overline{V}) \in \mathrm{Var}_{\overline{K}}^{\mathrm{ss}}$, the natural map*

$$\mathrm{R}\Gamma_{\mathrm{HK}}(V, \overline{V}) \to \mathrm{R}\Gamma_{\mathrm{HK}}(V)$$

is a quasi-isomorphism.

(ii) [**?**, Theorem 3.18] *For any $(V, \overline{V}) \in \mathrm{Var}_{K}^{\mathrm{ss}}$, the natural map*

$$\mathrm{R}\Gamma_{\mathrm{HK}}(V, \overline{V}) \to \mathrm{R}\Gamma_{\mathrm{HK}}(V)$$

is a quasi-isomorphism.

Remark 3.2. According to [**?**, Theorem 3.18], the property also holds for log smooth schemes over $\mathcal{O}_K^{\times}$. We will not use this fact in the following sections.

3.1.2. *Algebraic Hyodo-Kato cohomology with compact support.* We define the Hyodo-Kato cohomology with compact support in this section.

We first recall some basic definitions and properties of compactifications. For a morphism $X \to Y$ of schemes, we say X has a compactification over Y if there exists a quasi-compact open immersion $X \to \overline{X}$ such that $\overline{X}$ is proper over Y. For $X \to Y$ separated and of finite type, such a compactification always exists. The category of compactifications of X over Y is cofiltered, which means that for any two compactifications $X \to \overline{X_1}$ and $X \to \overline{X_2}$, there exists a compactification $X \to \overline{X}$ and morphisms $\overline{X_1} \leftarrow \overline{X} \to \overline{X_2}$ over Y such that both morphisms are isomorphisms on the open subset X. Moreover, given a morphism $X \to Y$ of algebraic varieties over a field L and a compactification $X \to \overline{X}$ over L, one can construct a compactification $X \to \overline{X'}$ over Y by taking $\overline{X'}$ to be the scheme-theoretic image of $X \to \overline{X} \times_L Y$.

Definition 3.3. Let X be an algebraic variety, and let $X \hookrightarrow \overline{X}$ be a compactification of X over L. We define the Hyodo-Kato cohomology with compact support of X to be

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(X) := [\mathrm{R}\Gamma_{\mathrm{HK}}(\overline{X}) \to \mathrm{R}\Gamma_{\mathrm{HK}}(\overline{X} - X)].$$

Since Hyodo-Kato cohomology satisfies h-descent, one checks easily that $\mathrm{R}\Gamma_{\mathrm{HK,c}}(X)$ is well-defined: indeed, suppose we have two compactifications of X over L: $X \hookrightarrow \overline{X_1}$ and $X \hookrightarrow \overline{X_2}$, then we have a compactification $X \to \overline{X}$ and morphisms $\overline{X_1} \leftarrow \overline{X} \to \overline{X_2}$. Then we have an abstract blow-up square:

$$\begin{array}{ccc} \overline{X} - X & \longrightarrow & \overline{X} \\ \downarrow & & \downarrow \\ \overline{X_i} - X & \longrightarrow & \overline{X_i} \end{array}$$

for $i = 1, 2$. This is in fact an éh-covering in the sense of [**?**].

Remark 3.4. For a general cohomology theory satisfying éh-descent, one can similarly define cohomology with compact support.

Proposition 3.5. *Let X be an algebraic variety, and let $U \hookrightarrow X$ be an open immersion. Then we have*

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(U) \simeq [\mathrm{R}\Gamma_{\mathrm{HK,c}}(X) \to \mathrm{R}\Gamma_{\mathrm{HK,c}}(X - U)].$$

Proof. Choose a compactification $X \hookrightarrow \overline{X}$ of X. Then $\overline{X}$ is also a compactification of U, and $\overline{X} - U$ is a compactification of $X - U$. Then the proposition follows from the definition of compactly supported cohomology for X, U, and $X - U$. □

Proposition 3.6. *Hyodo-Kato cohomology with compact support is contravariant for proper morphisms and covariant for open immersions.*

Proof. If $U \hookrightarrow X$ is an open immersion, as above we choose a compactification $X \hookrightarrow \overline{X}$ of X. Then the map $\mathrm{R}\Gamma_{\mathrm{HK}}(\overline{X} - U) \to \mathrm{R}\Gamma_{\mathrm{HK}}(\overline{X} - X)$ induces $\mathrm{R}\Gamma_{\mathrm{HK,c}}(U) \to \mathrm{R}\Gamma_{\mathrm{HK,c}}(X)$.

If $Y \to X$ is a proper morphism, choose a compactification $X \hookrightarrow \overline{X}$ of X and a compactification $Y \hookrightarrow \overline{Y'}$ of Y. Denote by $\overline{Y}$ the closure of the image of the canonical morphism $Y \to \overline{Y'} \times_L \overline{X}$. We have a Cartesian square

$$\begin{array}{ccc} Y & \hookrightarrow & \overline{Y} \\ \downarrow & & \downarrow \\ X & \hookrightarrow & \overline{X}, \end{array}$$

because the map from Y to the preimage of X in $\overline{Y}$ is a proper open immersion with dense image, hence an isomorphism. Then $\mathrm{R}\Gamma_{\mathrm{HK,c}}(X) \to \mathrm{R}\Gamma_{\mathrm{HK,c}}(Y)$ is given by the following commutative square

$$\begin{array}{ccc} \overline{Y} - Y & \hookrightarrow & \overline{Y} \\ \downarrow & & \downarrow \\ \overline{X} - X & \hookrightarrow & \overline{X}. \end{array}$$

□

3.1.3. *The Hyodo-Kato realization functor.* Hyodo-Kato cohomology with compact support is also defined in [**?**] by constructing the Hyodo-Kato realization functor and then applying the six-functor formalism for Voevodsky's motives. Here we briefly summarize this construction. It will give us the pairing and Poincaré duality.

Let $DM(\overline{K},\mathbb{Q}_p)$ be the triangulated category of Voevodsky's mixed motives with $\mathbb{Q}_p$-coefficients, and let M_{gm} be the natural functor

$$M_{gm} : \mathrm{Sch}\,/\overline{K} \to DM(\overline{K},\mathbb{Q}_p).$$

We refer the reader to [**?**] for the definition and properties of Voevodsky's motives. According to [**?**], there exists a Hyodo-Kato realization functor

$$\mathrm{R}\Gamma_{\mathrm{HK}} : DM(\overline{K},\mathbb{Q}_p) \to D(\mathrm{Mod}^{\mathrm{solid}}_{F^{\mathrm{nr}}}),$$

such that its composition with M_{gm} gives the geometric Hyodo-Kato cohomology of Beilinson, i.e., for $X \in \mathrm{Sch}\,/\overline{K}$, we have $\mathrm{R}\Gamma_{\mathrm{HK}}(M_{gm}(X)) \simeq \mathrm{R}\Gamma_{\mathrm{HK}}(X)$. The realization functor also gives us the Gysin sequence for smooth pairs.

Proposition 3.7. *If X is a smooth variety over $\overline{K}$ and $Z \subset X$ is a smooth closed subvariety, set $U := X - Z$. We have*

$$\mathrm{R}\Gamma_{\mathrm{HK}}(Z)\{-1\}[-2] \simeq [\mathrm{R}\Gamma_{\mathrm{HK}}(X) \to \mathrm{R}\Gamma_{\mathrm{HK}}(U)].$$

Proof. This follows from the distinguished triangle of [**?**, Proposition 3.5.4]. □

There is also a functor

$$M^c_{gm} : \mathrm{Sch}\,/\overline{K} \to DM(\overline{K},\mathbb{Q}_p)$$

from the category of schemes of finite type over $\overline{K}$ to $DM(\overline{K},\mathbb{Q}_p)$, which sends $X \in \mathrm{Sch}\,/\overline{K}$ to its associated motive with compact support. We define the Hyodo-Kato cohomology with compact support of X by

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(X) := \mathrm{R}\Gamma_{\mathrm{HK}}(M^c_{gm}(X)).$$

Lemma 3.8. *The definition of Hyodo-Kato cohomology with compact support in* [**?**] *agrees with Definition* **??**.

Proof. This follows from the distinguished triangle of [**?**, Proposition 4.1.5]. □

Thanks to the six-functor formalism for Voevodsky's mixed motives (see, for example, [**?**]), one can construct a pairing for smooth varieties X:

$$\mathrm{R}\Gamma_{\mathrm{HK}}(X) \otimes \mathrm{R}\Gamma_{\mathrm{HK,c}}(X) \to \mathrm{R}\Gamma_{\mathrm{HK,c}}(X),$$

and the Hyodo-Kato realization functor gives us Poincaré duality for algebraic varieties as follows.

Theorem 3.9. *Let X be a smooth algebraic variety over $\overline{K}$ of dimension d. Then:*

(i) There is a natural trace map

$$\mathrm{tr}_{\mathrm{HK}} : \mathrm{R}\Gamma_{\mathrm{HK,c}}(X)[2d] \to F^{\mathrm{nr}}\{-d\},$$

compatible with the Hyodo-Kato morphism.

(ii) The pairing

$$\mathrm{R}\Gamma_{\mathrm{HK}}(X) \otimes^L \mathrm{R}\Gamma_{\mathrm{HK,c}}(X)[2d] \to \mathrm{R}\Gamma_{\mathrm{HK,c}}(X)[2d] \to F^{\mathrm{nr}}\{-d\}$$

is a perfect duality of solid F^{nr}-vector spaces, i.e., we have induced isomorphisms in $D(\mathrm{Mod}_{F^{\mathrm{nr}}}^{\mathrm{solid}})$:

$$H^i_{\mathrm{HK}}(X) \simeq \mathrm{Hom}_{F^{\mathrm{nr}}}(H^{2d-i}_{\mathrm{HK,c}}(X), F^{\mathrm{nr}}\{-d\}),$$

$$H^i_{\mathrm{HK,c}}(X) \simeq \mathrm{Hom}_{F^{\mathrm{nr}}}(H^{2d-i}_{\mathrm{HK}}(X), F^{\mathrm{nr}}\{-d\}).$$

3.2. **Hyodo-Kato cohomology for analytic varieties.** We review the definitions and properties of Hyodo-Kato cohomology (with compact support) of analytic varieties.

3.2.1. *Definition and properties.* We briefly summarize the definition and basic properties of Hyodo-Kato cohomology discussed in [**?**] and [**?**].

(1) Arithmetic setup: Here we define Hyodo-Kato cohomology of smooth rigid analytic varieties over K. Suppose $\mathscr{X} \in \mathscr{M}_K^{\mathrm{ss}}$, as described before Proposition **??**. Let $\mathscr{A}_{\mathrm{HK}}$ be the éh-sheafification of the presheaf

$$\mathscr{X} \mapsto \mathrm{R}\Gamma_{\mathrm{HK}}(\mathscr{X}_0) := \mathrm{R}\Gamma_{\mathrm{cris}}(\mathscr{X}_0/\mathcal{O}^0_{F_L})_{\mathbb{Q}_p}$$

with values in $D(\mathrm{Mod}_F^{\mathrm{solid}})$. Here the condensed structure of $\mathrm{R}\Gamma_{\mathrm{cris}}(\mathscr{X}_0/\mathcal{O}^0_{F_L})$ comes from the inverse limit of the condensed abelian groups $\varprojlim \mathrm{R}\Gamma_{\mathrm{cris}}(\mathscr{X}_0/\mathcal{O}^0_{F_{L,n}})$. The sheaf $\mathscr{A}_{\mathrm{HK}}$ is an éh-sheaf of dg solid F-algebras on $\mathscr{M}_K$ equipped with a Frobenius operator ϕ and a monodromy operator N such that $N\phi = p\phi N$. For a rigid analytic variety $X \in \mathrm{Rig}_K$, we define the Hyodo-Kato cohomology of X by $\mathrm{R}\Gamma_{\mathrm{HK}}(X) := \mathrm{R}\Gamma(X, \mathscr{A}_{\mathrm{HK}})$. By [**?**], there exists a Hyodo-Kato morphism

$$\iota_{\mathrm{HK}} : \mathrm{R}\Gamma_{\mathrm{HK}}(X) \to \mathrm{R}\Gamma_{\mathrm{dR}}(X).$$

(2) Geometric setup: Suppose now that $\mathscr{X} \in \mathscr{M}_C^{\mathrm{b,ss}}$ is a basic semistable formal model. By definition, $f : \mathscr{X} \to \mathrm{Spf}(\mathcal{O}_C)^\times$ is the base change of a semistable formal model $\theta_L : \mathscr{X}_L \to \mathrm{Spf}(\mathcal{O}_L)^\times$ for a finite field extension L/K. The data $(L, \theta_L, \mathscr{X}_L)$ form a filtered set Σ. Note that for a morphism $(L, \theta_L, \mathscr{X}_L) \to (L', \theta_{L'}, \mathscr{X}'_{L'})$ in Σ, the base change theorem tells us

$$\mathrm{R}\Gamma_{\mathrm{HK}}(\mathscr{X}'_{L',0}) \otimes_{F'_L} F_L \simeq \mathrm{R}\Gamma_{\mathrm{HK}}(\mathscr{X}_{L,0}).$$

Let $\mathscr{A}_{\mathrm{HK}}$ be the éh-sheafification of the presheaf

$$\mathscr{X} \mapsto \mathrm{R}\Gamma_{\mathrm{HK}}(\mathscr{X}_0) := \varinjlim_L \mathrm{R}\Gamma_{\mathrm{HK}}(\mathscr{X}_{L,0}),$$

where the filtered colimit is indexed by Σ. This is an éh-sheaf of dg solid F^{nr}-algebras on Rig_C equipped with a Frobenius operator ϕ and a monodromy operator N such that $N\phi = p\phi N$. For a rigid analytic variety $X \in \mathrm{Rig}_C$, we define the Hyodo-Kato cohomology of X by $\mathrm{R}\Gamma_{\mathrm{HK}}(X) := \mathrm{R}\Gamma(X, \mathscr{A}_{\mathrm{HK}})$.

We have the following local-global compatibility.

Proposition 3.10. [**?**, Propositions 4.11 and 4.23] *(i) For a semistable formal scheme $\mathscr{X}$ over $\mathcal{O}_K$, the natural morphism*

$$\mathrm{R}\Gamma_{\mathrm{HK}}(\mathscr{X}_1) \to \mathrm{R}\Gamma_{\mathrm{HK}}(\mathscr{X}_K)$$

in $D(\mathrm{Mod}_F^{\mathrm{solid}})$ is a strict quasi-isomorphism.

(ii) For $\mathscr{X} \in \mathscr{M}_C^{\mathrm{ss,b}}$, the natural morphism

$$\mathrm{R}\Gamma_{\mathrm{HK}}(\mathscr{X}_1) \to \mathrm{R}\Gamma_{\mathrm{HK}}(\mathscr{X}_C)$$

in $D(\mathrm{Mod}_{F^{\mathrm{nr}}}^{\mathrm{solid}})$ is a strict quasi-isomorphism.

In the geometric setup, we have the Hyodo-Kato isomorphism.

The following Hyodo-Kato isomorphism follows from [**?**, Theorem 4.6] and [**?**, Theorem 2.3.14].

Proposition 3.11. *Suppose X is a dagger variety over C. Then we have a natural isomorphism in $D(\mathrm{Mod}_C^{\mathrm{solid}})$:*

$$\mathrm{R}\Gamma_{\mathrm{HK}}(X) \otimes^{\mathrm{L}\blacksquare}_{F^{\mathrm{nr}}} C \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{dR}}(X).$$

3.2.2. *Compactly supported Hyodo-Kato cohomology for rigid analytic varieties.* We review the definition of compactly supported Hyodo-Kato cohomology for rigid analytic varieties given in [**?**, Section 3.2].

Definition 3.12. Let X be a rigid analytic variety over $L = K$ or C. The compactly supported Hyodo-Kato cohomology of X is defined to be

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(X) := \varinjlim_{U \subset X} [\mathrm{R}\Gamma_{\mathrm{HK}}(X) \to \mathrm{R}\Gamma_{\mathrm{HK}}(X - U)]$$

in $D(\mathrm{Mod}_{L_F}^{\mathrm{solid}})$, where the filtered colimit is indexed by quasi-compact opens U in X, and $L_F = F$ when $L = K$ and $L_F = F^{\mathrm{nr}}$ when $L = C$.

We have similar definitions for compactly supported de Rham, B_{dR}^+, étale, syntomic and pro-étale cohomology for rigid analytic varieties. All these cohomology groups with compact support are contravariant for proper morphisms and covariant for open immersions.

Remark 3.13. One should be careful with the terminology used. When X is partially proper, the definition of compactly supported pro-étale cohomology coincides with Huber's definition of compactly supported étale cohomology in [**?**] (and [**?**] for p-adic cohomology). See [**?**, Section 2] for more details.

3.2.3. *Compactly supported Hyodo-Kato cohomology for dagger analytic varieties.* Now suppose X is a dagger variety over $L = K$ or C. We first assume that X is dagger affinoid. According to [**?**], we can take a presentation $\{X_k\}_{k\geq 1}$ of X, where all X_k are smooth affinoid rigid analytic varieties over L such that $X_1 \ni X_2 \ni \cdots$ and their intersection is X. Denote by $\mathscr{A}_{\mathrm{HK}}$ the analytic sheafification of the presheaf

$$X \mapsto \mathrm{R}\Gamma^{\natural}_{\mathrm{HK,c}}(X) := \mathrm{colim}_k[\mathrm{R}\Gamma_{\mathrm{HK}}(X_k) \to \mathrm{R}\Gamma_{\mathrm{HK}}(X_k - X)].$$

For a general X, define the overconvergent compactly supported Hyodo-Kato cohomology by

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(X) := \mathrm{R}\Gamma(X, \mathscr{A}_{\mathrm{HK}}).$$

This definition satisfies local-global compatibility.

Proposition 3.14. [**?**, Corollary 4.16] *If X is dagger affinoid, then we have a natural quasi-isomorphism*

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma^{\natural}_{\mathrm{HK,c}}(X).$$

We have similar definitions and properties for compactly supported de Rham, B^+_{dR}, syntomic and pro-étale cohomology for dagger analytic varieties.

When X is partially proper, we can compare the two definitions as expected.

Proposition 3.15. [**?**, Proposition 4.18] *Suppose moreover that X is partially proper. Then we have a natural quasi-isomorphism*

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK,c}}(\widehat{X}).$$

The same statement also holds for compactly supported de Rham, B^+_{dR}, étale and pro-étale cohomology.

3.2.4. *Galois descent.* In the arithmetic setup, one can recover Hyodo-Kato cohomology by Galois descent.

Proposition 3.16. *Suppose that X is a dagger variety over K, and set $G_K := \mathrm{Gal}(\overline{K}/K)$. Then the natural projection $X_C \to X$ induces natural isomorphisms in $D(\mathrm{Mod}^{\mathrm{solid}}_{\mathbb{Q}_p})$:*

$$\mathrm{R}\Gamma_{\mathrm{HK}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(X_C)^{G_K}, \qquad \mathrm{R}\Gamma_{\mathrm{HK,c}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK,c}}(X_C)^{G_K}.$$

Proof. We may assume X is dagger affinoid, as solid G_K-cohomology commutes with limits and also filtered colimits since G_K is profinite. For Hyodo-Kato cohomology, this follows from [**?**, Remark 5.21] and the finiteness of Hyodo-Kato cohomology. The claim for Hyodo-Kato cohomology with compact support then follows. □

4. Geometric properties

As outlined in the introduction, this section is devoted to proving some geometric properties of analytic varieties, mainly dagger varieties. This includes the Mayer-Vietoris property, Poincaré duality, the open-closed exact sequence and the Gysin isomorphism.

4.1. **Mayer-Vietoris property.** Using the results from [**?**], we construct several Mayer-Vietoris exact sequences for rigid analytic and dagger varieties. We start with an abstract definition of the Mayer-Vietoris property.

Definition 4.1. Let $\mathcal{C}$ be a site and $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$ (resp. $F : \mathcal{C} \to \mathcal{D}$) be a presheaf (resp. precosheaf) valued in a stable ∞-category $\mathcal{D}$. For a commutative square

$$\begin{array}{ccc} Y' & \longrightarrow & X' \\ \downarrow & & \downarrow \\ Y & \longrightarrow & X \end{array}$$

we will say that F satisfies the Mayer-Vietoris property for this square if F sends it to a cartesian square. In this case, we will also call this square F-acyclic.

Example 4.2. Let $\mathcal{C} = \mathrm{Sch}_\tau$ (or $\mathrm{Rig}_\tau, \mathrm{Rig}^\dagger_\tau$,...) with a Grothendieck topology τ (of suitably bounded cardinality if τ is large), $\mathcal{D} = \mathcal{D}(\mathrm{CondAb})$, $F = \mathrm{R}\Gamma(-, \mathcal{F})$ for a condensed abelian sheaf

$\mathcal{F}$ on X_τ. Then the square

$$\begin{array}{ccc} U \cap V & \longrightarrow & V \\ \downarrow & & \downarrow \\ U & \longrightarrow & X \end{array}$$

is F-acyclic for any U, V open in X with $X = U \cup V$. This is the usual Mayer-Vietoris sequence. Similarly, for $\mathcal{C} = \mathrm{Sch}_\tau$ or $\mathrm{Rig}^\dagger_\tau$, the square is also F-acyclic for the cosheaf $F = \mathrm{R}\Gamma_c(-, \mathcal{F})$.

Example 4.3. [**?**] implies that, for a commutative ring Λ, $F = \mathrm{R}\Gamma_{\text{ét}}(-, \underline{\Lambda}) : (\mathrm{Sch}/k)^{op} \to \mathcal{D}(\mathrm{Ab})$ has the Mayer-Vietoris property for abstract blowup squares. An abstract blowup square means that $Y \hookrightarrow X$ is a closed immersion, $f : X' \to X$ is proper and surjective and induces an isomorphism $X' - f^{-1}(Y) \to X - Y$, and $Y' = X' \times_X Y$. We will prove similar results in the rigid analytic/dagger settings.

Example 4.4. According to [**?**] and [**?**], $F = \mathrm{R}\Gamma_{\mathrm{dR}}(-)$ or $F = \mathrm{R}\Gamma_{\mathrm{HK}}(-)$ has the Mayer-Vietoris property for proper surjective cartesian squares, i.e. $Y' = X' \times_X Y$ and $X' \to X$ is proper and surjective.

Now we will state the main result of this section.

Proposition 4.5. *Suppose $\mathcal{C} = \mathrm{Rig}_L$ or $\mathrm{Rig}^\dagger_L$, $L = K$ or C. Let $X \in \mathcal{C}$, let $Y \subset X$ be a nowhere dense closed analytic subspace, and set $Y' = X' \times_X Y$. Suppose that the square*

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

satisfies one of the following two conditions:

1. *$X' \to X$ is a blowup along Y.*
2. *X is quasi-compact, Y is an irreducible component of X, and X' is the union of all the other irreducible components of X.*

For the functors $F = \mathrm{R}\Gamma_{\text{proét}}(-, \mathbb{Q}_p)$ or $\mathrm{R}\Gamma_{\text{ét}}(-, \mathbb{Q}_p)$ or $\mathrm{R}\Gamma_{\text{éh}}(-, \mathcal{F})$ or $\mathrm{R}\Gamma_{\mathrm{dR}}(-)$ or $\mathrm{R}\Gamma(-/B^+_{\mathrm{dR}})$ or $\mathrm{R}\Gamma_{\mathrm{HK}}(-)$ (only for $\mathcal{C} = \mathrm{Rig}^\dagger_L$ in this case) from $\mathcal{C}^{\mathrm{op}}$ to $D(\mathrm{Mod}^{\mathrm{solid}}_{\mathbb{Q}_p})$, the above commutative square is F-acyclic.

Remark 4.6. After covering X by open (dagger) affinoids, since limits and colimits of cartesian squares are still cartesian, we may assume that X is affinoid or dagger affinoid in each situation (except the argument for étale cohomology).

Proof. We assume first that X is affinoid.

- éh and de Rham cohomology: This follows from [**?**, Proposition 5.1.4].
- étale and pro-étale cohomology: The sheaf $X \mapsto \mathrm{R}\Gamma(X_{\text{ét}}, \Lambda)$ is a v-sheaf, hence an éh-sheaf over $\mathrm{Rig}_{\text{éh}}$. Therefore, $\mathrm{R}\Gamma(X_{\text{ét}}, \Lambda) = \mathrm{R}\Gamma(X_{\text{éh}}, \Lambda)$ for any commutative ring Λ. After taking limits we get the Mayer-Vietoris property for $\mathrm{R}\Gamma(X_{\text{ét}}, \mathbb{Z}_p)$; tensoring with $\mathbb{Q}_p$, we get the Mayer-Vietoris property for $\mathrm{R}\Gamma(X_{\text{ét}}, \mathbb{Q}_p)$. Since X is quasi-compact, $\mathrm{R}\Gamma(X_{\text{ét}}, \mathbb{Q}_p) = \mathrm{R}\Gamma(X_{\text{proét}}, \mathbb{Q}_p)$ and therefore the Mayer-Vietoris property also holds for $\mathrm{R}\Gamma(X_{\text{proét}}, \mathbb{Q}_p)$.

Now we assume X is dagger affinoid.

- éh and de Rham cohomology: The proof is the same as in [**?**, Proposition 5.1.4] except that here we need to use the overconvergent setup.
- étale cohomology: By [**?**], $\mathrm{R}\Gamma(X_{\text{ét}},\Lambda)\simeq\mathrm{R}\Gamma(\widehat{X}_{\text{ét}},\Lambda)$ for any commutative ring Λ, and the Mayer-Vietoris property follows from the affinoid case.
- pro-étale cohomology: We use the presentation of an affinoid dagger variety. Take a presentation $\{X_h\}$ of X, and we may assume X_1 is small enough so that Y can be extended to a closed subvariety $Y_1\subset X_1$ (because X is affinoid, Y is defined by a finite number of overconvergent power series that extend to a slightly larger neighborhood). Therefore, we may assume X has a presentation $\{X_h\}$ that induces a presentation $\{Y_h\}$ of Y such that $Y_h\hookrightarrow X_h$ are closed immersions.

  We can then construct $X'_h=\mathrm{Bl}_{Y_h}(X_h)$, the blowup of X_h along Y_h, and set $Y'_h=X'_h\times_{X_h}Y_h$. We note that $X'_h=X'_1\times_{X_1}X_h$ for any h. According to the definition of pro-étale cohomology of dagger varieties, $\mathrm{R}\Gamma(X_{\text{proét}},\mathbb{Q}_p)=\mathrm{colim}_h\,\mathrm{R}\Gamma(X_{h,\text{proét}},\mathbb{Q}_p),\mathrm{R}\Gamma(Y_{\text{proét}},\mathbb{Q}_p)\simeq\mathrm{colim}_h\,\mathrm{R}\Gamma(Y_{h,\text{proét}},\mathbb{Q}_p)$. We want to prove that

  $$\mathrm{R}\Gamma(X'_{\text{proét}},\mathbb{Q}_p)\simeq\mathrm{colim}_h\,\mathrm{R}\Gamma(X'_{h,\text{proét}},\mathbb{Q}_p),\mathrm{R}\Gamma(Y'_{\text{proét}},\mathbb{Q}_p)=\mathrm{colim}_h\,\mathrm{R}\Gamma(Y'_{h,\text{proét}},\mathbb{Q}_p).$$

  According to the construction of blowups, if we assume $X_h=\mathrm{Sp}(A_h)$ and $X=\mathrm{Sp}^\dagger(A)$, we can find a finite dagger affinoid hypercovering of X' by $U^\bullet$ such that locally they are disjoint unions of $\mathrm{Spec}(A_{(a)})$ (relative Spec, see [**?**]), and we can construct hypercoverings of X'_h by $U^\bullet_h$ in a compatible way (which means locally they are disjoint unions of $\mathrm{Spec}(A_{h,(a)})$). Therefore,

  $$\mathrm{R}\Gamma(X'_{\text{proét}},\mathbb{Q}_p)\simeq\mathrm{R}\Gamma(U^\bullet_{\text{proét}},\mathbb{Q}_p)\simeq\mathrm{colim}_h\,\mathrm{R}\Gamma(U^\bullet_{h,\text{proét}},\mathbb{Q}_p)\simeq\mathrm{colim}_h\,\mathrm{R}\Gamma(X'_{h,\text{proét}},\mathbb{Q}_p),$$

  where the middle isomorphism follows from the fact that colimits commute with finite limits. The same argument also shows that $\mathrm{R}\Gamma(Y'_{\text{proét}},\mathbb{Q}_p)\simeq\mathrm{colim}_h\,\mathrm{R}\Gamma(Y'_{h,\text{proét}},\mathbb{Q}_p)$.
- Hyodo-Kato cohomology: We prove the claim for $L=C$. We know that $\mathrm{R}\Gamma_{\mathrm{HK}}(X)\otimes^{\blacksquare}_{F^{\mathrm{nr}}}C\simeq\mathrm{R}\Gamma_{\mathrm{dR}}(X)$. Write $C\simeq F^{\mathrm{nr}}\oplus W$ for some F^{nr}-vector space W. Since $\mathrm{R}\Gamma_{\mathrm{dR}}(X)$ has the Mayer-Vietoris property, $\mathrm{R}\Gamma_{\mathrm{HK}}(X)$, as a direct summand of $\mathrm{R}\Gamma_{\mathrm{dR}}(X)$ in $D(\mathrm{Mod}^{\mathrm{solid}}_{\mathbb{Q}_p})$, must also have the Mayer-Vietoris property.

  When $L=K$, the claim follows from the case for $L=C$, the finiteness of Hyodo-Kato cohomology (Proposition **??**), and Galois descent.

□

The above proof for dagger varieties also yields the Mayer-Vietoris property for cohomology with compact support on dagger varieties.

Proposition 4.7. *The above proposition also holds for $\mathcal{C}=\mathrm{Rig}^\dagger_L$, $L=K$ or C, and $F=\mathrm{R}\Gamma_{\text{proét,c}}(-,\mathbb{Q}_p)$ or $\mathrm{R}\Gamma_{\mathrm{dR,c}}(-/B^+_{\mathrm{dR}})$ or $\mathrm{R}\Gamma_{\mathrm{HK,c}}(-)$.*

Proof. By assuming that X is dagger affinoid and covering X' with dagger affinoids as above, we deduce the proposition from the cosheaf condition and the above Mayer-Vietoris property for the corresponding cohomology (without compact support). □

4.2. **The open-closed exact sequence.** A characteristic property of cohomology with compact support is the existence of an open-closed long exact sequence. For example, for étale cohomology with compact support, let X be a partially proper rigid analytic variety, let $D \hookrightarrow X$ be a Zariski closed immersion, and let $U := X - D \hookrightarrow X$ be the complementary open immersion. We have the following proposition, which follows from [**?**, Remark 5.1.1].

Proposition 4.8. *We have*

$$\mathrm{R}\Gamma_{\text{ét,c}}(U, \mathbb{Q}_p) \simeq [\mathrm{R}\Gamma_{\text{ét,c}}(X, \mathbb{Q}_p) \to \mathrm{R}\Gamma_{\text{ét,c}}(D, \mathbb{Q}_p)].$$

In this section, we will prove that our definition of cohomology with compact support admits an open-closed exact sequence when X is smooth and either X is proper, Z is smooth, or Z is a strict normal crossings divisor.

Let $L = K$ or C, and suppose X is a smooth dagger variety over L. Let $i : Z \hookrightarrow X$ be a Zariski closed immersion, and set $U = X - Z$.

Proposition 4.9. *Assume moreover that Z is smooth, or Z is a strict normal crossings divisor. Then we have*

$$\mathrm{R}\Gamma_{\mathrm{dR,c}}(U) \simeq [\mathrm{R}\Gamma_{\mathrm{dR,c}}(X) \to \mathrm{R}\Gamma_{\mathrm{dR,c}}(Z)],$$
$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(U) \simeq [\mathrm{R}\Gamma_{\mathrm{HK,c}}(X) \to \mathrm{R}\Gamma_{\mathrm{HK,c}}(Z)].$$

Proof. This problem is local, so we can assume $X = \mathrm{Sp}(A)$ is a smooth dagger affinoid variety. We prove the claim for de Rham cohomology first.

Choose a representation $A = L\langle x_1, ..., x_n\rangle^\dagger / I$, and assume $Z = \mathrm{Sp}(L\langle x_1, ..., x_n\rangle^\dagger / J)$. By definition we can find a $\delta > 1$ such that $I \subset J \subset T_n(\delta) = L\left\langle \delta^{-1}x_1, ..., \delta^{-1}x_n\right\rangle$. By shrinking δ we can assume $I \subset J \subset W_n(\delta) = L\left\langle \delta^{-1}x_1, ..., \delta^{-1}x_n\right\rangle^\dagger$. Take a decreasing sequence $\delta_1 = \delta, \delta_h \to 1$. Define $X_h = \mathrm{Sp}(W_n(\delta_h)/IW_n(\delta_h)), Z_h = \mathrm{Sp}(W_n(\delta_h)/JW_n(\delta_h))$. Moreover, choose generators $J = (f_1, ..., f_m)$. For any $\rho \geq 1$, write

$$B_k(\rho) = L\left\langle \rho^{-1}x_1, ..., \rho^{-1}x_n, \omega^{-k}f_1, ..., \omega^{-k}f_m\right\rangle^\dagger.$$

We can define a decreasing sequence of neighborhoods of Z_h (resp. Z) by $V_{k,h} = \mathrm{Sp}(B_k(\delta_h)/IB_k(\delta_h))$ (resp. $V_k = \mathrm{Sp}(B_k(1)/IB_k(1)) = V_{k,h} \cap X$).

Lemma 4.10. *We have natural isomorphisms*

$$\mathrm{R}\Gamma_{\mathrm{dR}}(Z_1) \simeq \mathrm{colim}_k \mathrm{R}\Gamma_{\mathrm{dR}}(V_{k,1}),$$
$$\mathrm{R}\Gamma_{\mathrm{dR}}(Z_1 - Z) \simeq \mathrm{colim}_k \mathrm{R}\Gamma_{\mathrm{dR}}(V_{k,1} - (V_{k,1} \cap X)).$$

Proof. The claims are given by the definition of de Rham cohomology by Elmar Grosse-Klönne, see Appendix **??**. For the first isomorphism, note that $V_{k,1}$ is a cofinal system of the set of admissible open neighborhoods $U \subset X_1$ of Z_1. Thus, by Theorem **??**, writing $j_k : V_{k,1} \hookrightarrow X_1$, we obtain

$$\mathrm{colim}_k \mathrm{R}\Gamma_{\mathrm{dR}}(V_{k,1}) \simeq \mathrm{colim}_k \mathrm{R}\Gamma(X_1, Rj_{k,*}\Omega^\bullet|_{V_{k,1}}) \simeq \mathrm{colim}_k \mathrm{R}\Gamma(X_1, j_{k,*}\Omega^\bullet|_{V_{k,1}}) \simeq \mathrm{R}\Gamma_{\mathrm{dR}}(Z_1).$$

The second isomorphism is more subtle, but when Z is smooth, or Z is a strict normal crossings divisor, by [**?**, Theorem 1.18], we may further assume

$$X \simeq S \times \mathbb{B}^r \simeq S \times \mathrm{Sp}(L\langle x_1, ..., x_r\rangle),$$

$$X_h \simeq S_h \times \mathbb{B}(\delta_h)^r \simeq S_h \times \mathrm{Sp}(L\langle \delta_h^{-1}x_1, ..., \delta_h^{-1}x_r\rangle),$$

where S, S_h are smooth affinoids, and

$$Z_h \simeq S_h \simeq S_h \times \mathrm{Sp}(L\langle \delta_h^{-1}x_1, ..., \delta_h^{-1}x_r\rangle/(x_1, ..., x_r)) \hookrightarrow X_h$$

when Z is smooth, or

$$Z_h \simeq S_h \times \mathrm{Sp}(L\langle \delta_h^{-1}x_1, ..., \delta_h^{-1}x_r\rangle/(x_1x_2...x_r)) \hookrightarrow X_h$$

when Z is a strict normal crossings divisor. Let $j_{k,h} : V_{k,h} - (V_{k,h} \cap X) \hookrightarrow X_h - X$. When Z is smooth, we can let $V_{k,h} \simeq S_h \times \mathbb{B}(\omega^k)^r$ and the source of $j_{k,h}$ can be described as $(S_h - S) \times \mathbb{B}(\omega^k)^r$. By Theorem **??**,

$$\begin{aligned}
\mathrm{R}\Gamma_{\mathrm{dR}}(Z_1 - Z) &\simeq \mathrm{R}\Gamma(X_1 - X, \mathrm{colim}_k\, j_{k,1,*}\Omega^\bullet_{V_{k,1}-V_k}) \\
&\simeq R\mathrm{lim}_h\, \mathrm{R}\Gamma(X_1 - X_h^\circ, \mathrm{colim}_k\, j_{k,1,*}\Omega^\bullet_{V_{k,1}-V_k}|_{X_1-X_h^\circ}) \\
&\simeq R\mathrm{lim}_h\, \mathrm{colim}_k\, \mathrm{R}\Gamma_{\mathrm{dR}}(V_{k,1} - V_{k,h}^\circ) \\
&\simeq R\mathrm{lim}_h\, \mathrm{R}\Gamma_{\mathrm{dR}}(V_{k,1} - V_{k,h}^\circ) \simeq \mathrm{R}\Gamma_{\mathrm{dR}}(V_{k,1} - (V_{k,1} \cap X)).
\end{aligned}$$

When Z is a strict normal crossings divisor, one can use the éh-descent to reduce to the smooth case. □

By definition, we have

$$\mathrm{R}\Gamma_{\mathrm{dR,c}}(X) \simeq \mathrm{colim}_k[\mathrm{R}\Gamma_{\mathrm{dR}}(X_1) \to \mathrm{R}\Gamma_{\mathrm{dR}}((X_1 - X))].$$

We also have

$$\begin{aligned}
\mathrm{R}\Gamma_{\mathrm{dR,c}}(U) &\simeq \mathrm{colim}_k\, \mathrm{R}\Gamma_{\mathrm{dR,c}}(X - V_k) \\
&\simeq \mathrm{colim}_k[\mathrm{R}\Gamma_{\mathrm{dR}}(X_1) \to \mathrm{R}\Gamma_{\mathrm{dR}}((X_1 - X) \cup V_{k,1})]
\end{aligned}$$

Using the lemma above, we have

$$\begin{aligned}
\mathrm{R}\Gamma_{\mathrm{dR,c}}(Z) &\simeq [\mathrm{R}\Gamma_{\mathrm{dR}}(Z_1) \to \mathrm{R}\Gamma_{\mathrm{dR}}((Z_1 - Z))] \\
&\simeq \mathrm{colim}_k[\mathrm{R}\Gamma_{\mathrm{dR}}(V_{k,1}) \to \mathrm{R}\Gamma_{\mathrm{dR}}(V_{k,1} - (V_{k,1} \cap X))] \\
&\simeq \mathrm{colim}_k[\mathrm{R}\Gamma_{\mathrm{dR}}((X_1 - X) \cup V_{k,1}) \to \mathrm{R}\Gamma_{\mathrm{dR}}((X_1 - X))].
\end{aligned}$$

The last isomorphism follows from the excision square:

$$\begin{array}{ccc}
V_{k,1} - (V_{k,1} \cap X) & \longrightarrow & V_{k,1} \\
\downarrow & & \downarrow \\
X_1 - X & \longrightarrow & (X_1 - X) \cup V_{k,1}.
\end{array}$$

By combining the above three isomorphisms we get the desired proposition for de Rham cohomology.

If X is defined over $L = C$, the proof for Hyodo-Kato cohomology follows from the Hyodo-Kato isomorphism. Then for $L = K$ we use Galois descent. This concludes the proof. □

Remark 4.11. If Z is not Zariski closed, it seems that there is no reasonable way to define $\mathrm{R}\Gamma_{\mathrm{dR,c}}(X) \to \mathrm{R}\Gamma_{\mathrm{dR,c}}(Z)$.

Remark 4.12. One could begin by assuming the existence of a tubular neighborhood and proving the proposition through direct computation. Initially, we aimed to establish the proposition for any Z without extra assumptions. According to the proof, the primary issue for a general Z is that $Z_1 - Z$ is not quasi-compact, and we do not have a straightforward characterization of the de Rham cohomology as described by Grosse-Klönne. It remains uncertain whether the assumptions on Z can be relaxed. However, if X is proper, the definition of cohomology with compact support used in rigid analytic varieties allows us to remove the assumptions on Z through a different approach, as presented below.

Proposition 4.13. *If X is proper, we have*

$$\mathrm{R}\Gamma_{\mathrm{dR,c}}(U) \simeq [\mathrm{R}\Gamma_{\mathrm{dR}}(X) \to \mathrm{R}\Gamma_{\mathrm{dR}}(Z)],$$
$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(U) \simeq [\mathrm{R}\Gamma_{\mathrm{HK}}(X) \to \mathrm{R}\Gamma_{\mathrm{HK}}(Z)].$$

Proof. By excision,

$$\begin{aligned}\mathrm{R}\Gamma_{\mathrm{dR,c}}(U) &\simeq \mathrm{colim}_{W\in\Phi(U)}[\mathrm{R}\Gamma_{\mathrm{dR}}(U) \to \mathrm{R}\Gamma_{\mathrm{dR}}(U-W)]\\ &\simeq \mathrm{colim}_{W\in\Phi(U)}[\mathrm{R}\Gamma_{\mathrm{dR}}(X) \to \mathrm{R}\Gamma_{\mathrm{dR}}(X-W)].\end{aligned}$$

It suffices to show

$$\mathrm{colim}_{W\in\Phi(U)}\,\mathrm{R}\Gamma_{\mathrm{dR}}(X-W) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{dR}}(Z).$$

Cover X by finitely many affinoid dagger varieties $X_1, ..., X_m$, and let $Z_i = X_i \cap Z$. Write

$$X_i = \mathrm{Sp}(A_i) = \mathrm{Sp}(L\langle x_1, ..., x_n\rangle^\dagger / I_i),$$
$$X_i^\circ = \bigcup_{\delta\to 1} \mathrm{Sp}(L\left\langle \delta^{-1}x_1, ..., \delta^{-1}x_n\right\rangle^\dagger / I_i),$$

and assume Z_i is cut out by an ideal $J_i = (f_{i1}, ..., f_{il}) \subset A_i$ (we can fix a large l and a large n for all i). For any sufficiently small $\rho > 0$, define

$$V_i(\rho) = \{|f_{ij}(x)| \le \rho, j = 1, 2, ..., l\} = \mathrm{Sp}\left(A_i\left\langle \rho^{-1}f_{i1}, ..., \rho^{-1}f_{il}\right\rangle^\dagger\right).$$

Set $V(\rho) = V_1(\rho) \cup ... \cup V_m(\rho)$.

The proposition follows from the next lemma.

Lemma 4.14. *$V(\rho)$ is an admissible open subset of X. Let $j(\rho) : V(\rho) \hookrightarrow X$. Then for any locally free sheaf $\mathcal{F}$ of finite rank on $V(\rho)$, $R^i j(\rho)_*\mathcal{F} = 0$ for $i > 0$. Moreover, $\{V(\rho)\}_\rho$ is a cofinal system of the set of admissible open neighborhoods $U \subset X$ of Z, and*

$$\mathrm{colim}_{W\in\Phi(U)}\,\mathrm{R}\Gamma_{\mathrm{dR}}(X-W) \simeq \mathrm{colim}_{\rho\to 0}\,\mathrm{R}\Gamma_{\mathrm{dR}}(X - V(\rho)).$$

Proof. To prove $V(\rho)$ is admissible, it suffices to show $V(\rho) \cap X_i$ is admissible for each i. For this we use [**?**, 9.1.4, Corollary 4]. The second claim is clear. [**?**, Lemma 2.3] implies $\{V(\rho)\}_\rho$ is a cofinal system of the set of admissible open neighborhoods $U \subset X$ of Z. For the last statement, for any sufficiently small ρ define

$$W_i(\rho) = \bigcup_j \mathrm{Sp}\left(A_i\left\langle \rho f_{ij}^{-1}\right\rangle^\dagger\right) \subset X_i$$

and $W(\rho) = W_1(\rho) \cup ... \cup W_m(\rho)$. By the same reference, $\{W(\rho)\}_\rho$ is a cofinal system of $\Phi(U)$. It suffices to show that for any ρ, there exist $\rho'' \leq \rho' \leq \rho$, such that

$$X - V(\rho) \subset W(\rho') \subset X - V(\rho'').$$

For the left inclusion, we can take $\rho = \rho'$ by the construction of $W(\rho)$. For the right inclusion, the same reference ensures the existence of ρ''. □

To complete the proof, since X is proper,

$$\begin{aligned}\mathrm{colim}_{W\in\Phi(U)}\,\mathrm{R}\Gamma_{\mathrm{dR}}(X-W) &\simeq \mathrm{colim}_{\rho\to 0}\,\mathrm{R}\Gamma_{\mathrm{dR}}(X-V(\rho))\\ &\simeq \mathrm{colim}_{\rho\to 0}\,\mathrm{R}\Gamma(X, j(\rho)_*\Omega^\bullet_{X-V(\rho)})\\ &\simeq \mathrm{R}\Gamma(X, \mathrm{colim}_{\rho\to 0}\, j(\rho)_*\Omega^\bullet_{X-V(\rho)})\\ &\simeq \mathrm{R}\Gamma_{\mathrm{dR}}(Z).\end{aligned}$$

The last isomorphism follows from the above lemma and Theorem **??** from Appendix **??**. Finally, the claim for Hyodo-Kato cohomology follows from the Hyodo-Kato isomorphism (for $L = C$, then for $L = K$ we use Galois descent). □

4.3. **Poincaré duality.** In this section, following [**?**], we will review Poincaré duality for de Rham cohomology and Hyodo-Kato cohomology of smooth dagger varieties. We will compare Poincaré duality for de Rham cohomology with the one constructed in [**?**]. We will then deduce the Gysin sequence from the open-closed long exact sequence and Poincaré duality.

Poincaré duality for de Rham cohomology is formulated as follows.

Theorem 4.15. [**?**, Theorems 5.23 and 5.29] *Let Y be a partially proper smooth rigid analytic variety or a quasi-compact smooth dagger variety over $L = K$ or C of dimension d. Then there is a natural trace map*

$$\mathrm{tr}_{\mathrm{dR}} : \mathrm{R}\Gamma_{\mathrm{dR,c}}(Y)[2d] \to L,$$

that induces a perfect pairing

$$\mathrm{R}\Gamma_{\mathrm{dR}}(Y) \otimes^{L\blacksquare}_{L} \mathrm{R}\Gamma_{\mathrm{dR,c}}(Y)[2d] \to \mathrm{R}\Gamma_{\mathrm{dR,c}}(Y)[2d] \to L.$$

Moreover, the trace map $\mathrm{tr}_{\mathrm{dR}}$ is compatible with restrictions to open dagger subvarieties.

In [**?**], the authors also constructed Poincaré duality for de Rham cohomology of an almost proper smooth rigid analytic variety Y over K, i.e., Y can be written as $X - Z$, where X is a proper smooth rigid analytic variety defined over K and Z is Zariski closed in X. The main goal of this section is to compare these two constructions, which will be used when proving GAGA in the next section.

Remark 4.16. In [**?**], the authors also constructed Poincaré duality for étale cohomology of almost proper smooth rigid analytic varieties. We refer the reader to [**?**, Theorem 4.4.1] for the statements.

We begin with compactly supported de Rham cohomology defined in [**?**]. Let $Y = X - Z$, where X is a proper smooth rigid analytic variety defined over $L = K$ or C and Z is a strict normal crossings divisor in X. We endow X with the log structure induced by Z. Let $\mathcal{I}$ be the invertible ideal sheaf of $\mathcal{O}_X$ associated to the closed immersion $Z \hookrightarrow X$.

Definition 4.17. [?, Definition 3.1.1] The compactly supported de Rham cohomology of Y (by Lan-Liu-Zhu) is defined to be

$$\mathrm{R}\Gamma_{\mathrm{dR,LLZ,c}}(Y) := \mathrm{R}\Gamma(X, \mathcal{I} \otimes_{\mathcal{O}_X} \Omega^{\bullet,\log}_{X/L})$$

in $D(\mathrm{Mod}(L^{\mathrm{solid}}))$.

Remark 4.18. This definition also applies to algebraic varieties.

Lemma 4.19. *We have a natural (non-filtered) quasi-isomorphism*

$$\mathrm{R}\Gamma_{\mathrm{dR,LLZ,c}}(Y) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{dR,c}}(Y).$$

Proof. This follows from Proposition **??** and [?, Proposition 4.3.4]: write Z as the union of irreducible components $Z = Z_1 \cup ... \cup Z_m$ where $Z_1, ..., Z_m$ are smooth. Denote by $Z^\bullet$ the Čech nerve of the map $\coprod Z_i \to Z$. Then both definitions can be written as

$$\mathrm{R}\Gamma_{\mathrm{dR},*,\mathrm{c}}(Y) \simeq [\mathrm{R}\Gamma_{\mathrm{dR}}(X) \to \mathrm{R}\Gamma_{\mathrm{dR}}(Z^\bullet)],$$

where $* = \varnothing$ or LLZ. □

Now suppose X is defined over $L = K$. Denote by

$$\mathrm{tr}_{\mathrm{dR,LLZ}} : \mathrm{R}\Gamma_{\mathrm{dR,LLZ,c}}(Y)[2d] \to K$$

the trace map defined in [?, Theorem 4.2.7]. We want to compare $\mathrm{tr}_{\mathrm{dR,LLZ}}$ with $\mathrm{tr}_{\mathrm{dR}}$. Both constructions rely on Serre duality for coherent sheaves on rigid analytic varieties, which was studied in [?]. We begin with the definition of algebraic local cohomology for local rings. Let $(R, \mathfrak{m})$ be a noetherian local ring, M be a finitely generated R-module, and $\widetilde{M}$ be its associated coherent $\mathrm{Spec}(R)$-module. Set $X := \mathrm{Spec}(R)$ and $U := X - \{\mathfrak{m}\}$. Consider the functor

$$\Gamma_{\mathfrak{m}}(M) := \{a \in M | \exists t \in \mathbb{N} \text{ such that } \mathfrak{m}^t a = 0\}.$$

Then the local cohomology of M with support in $\mathfrak{m}$ is defined to be

$$H^i_{\mathfrak{m}}(M) := H^i(R\Gamma_{\mathfrak{m}}(M)).$$

Now we define the canonical residue map. Let $R = L[[x_1, ..., x_d]]$ be the ring of formal power series over a p-adic field L, and $\mathfrak{m} = (x_1, ..., x_d)$. Let $\Omega^\bullet_{R/L}$ denote the algebra of continuous differential forms of R/L. Then $\Omega^d_{R/L} \simeq L[[x_1, ..., x_d]]dx_1 \wedge dx_2 \wedge ... \wedge dx_d$. By [?], we have

$$H^d_{\mathfrak{m}}(\Omega^d_{R/L}) \simeq \left\{ \sum_\alpha a_\alpha x^\alpha dx_1 \wedge dx_2 \wedge ... \wedge dx_d, \alpha = (\alpha_1, ..., \alpha_d) \in \mathbb{Z}^d_{<0}, a_\alpha \in L \right\},$$

where the sum is finite. The residue map

$$\mathrm{res}_{(x)} : H^d_{\mathfrak{m}}(\Omega^d_{R/L}) \to L$$

is then given by

$$\sum_\alpha a_\alpha x^\alpha dx_1 \wedge dx_2 \wedge ... \wedge dx_d \mapsto a_{(-1,-1,...,-1)}.$$

By [?, Proposition 3.2.1], the residue map is independent of the choice of coordinates (x).

In general, let R be a complete regular local ring over L of dimension d, with maximal ideal $\mathfrak{m}$, such that $L' = R/\mathfrak{m}$ is a finite field extension of L. By choosing a regular system of parameters (x) of R, we have $R \simeq L'[[x_1, ..., x_d]]$. We can define the residue map

$$\mathrm{res}_{(x)} : H^d_{\mathfrak{m}}(\Omega^d_{R/L}) \to L$$

by composing the residue map above with the canonical trace map of the finite field extension L'/L. This is also independent of the choice of coordinates (x). Therefore, we will denote it by $\mathrm{res}_{\mathfrak{m}}$.

Now let Y be a rigid analytic variety over L, $\mathcal{F}$ a coherent sheaf on Y. Let $y \in Y$ be a classical point, i.e. defined by a finite field extension of L, and $\mathfrak{m}_y$ be the maximal ideal of the stalk $\mathcal{O}_{Y,y}$. Let $(R, \mathfrak{m}, M)$ be the $\mathfrak{m}_y$-adic completion of $(\mathcal{O}_{Y,y}, \mathfrak{m}_y, \mathcal{F}_y)$. According to [**?**, Lemma 4.1.1], there exists a canonical map

$$H^d_{\mathfrak{m}}(M) \to H^d(Y, \mathcal{F}). \tag{4.1}$$

If moreover Y is proper and smooth of pure dimension d and $\mathcal{F} = \Omega^d_{Y/L}$, then there exists a coherent trace map

$$\mathrm{tr}_{\mathrm{coh}} : H^d(Y, \Omega^d_{Y/L}) \to L,$$

whose pre-composition with (**??**) is the canonical residue map for any classical point y. $\mathrm{tr}_{\mathrm{coh}}$ is an isomorphism if Y is geometrically connected. The coherent trace map allows us to deduce Serre duality [**?**] for coherent complexes on proper smooth rigid analytic varieties.

$\mathrm{tr}_{\mathrm{dR,LLZ}}$ is then defined by the composition

$$H^{2d}_{\mathrm{dR,LLZ,c}}(Y) \simeq H^{d,d}_{\mathrm{Hodge,c}}(Y, \mathcal{O}_Y) := H^d(X, \mathcal{I} \otimes_{\mathcal{O}_X} \Omega^{d,\log}_X) \simeq H^d(X, \Omega^d_X) \to L,$$

where the first isomorphism follows from the degeneration of the Hodge–de Rham spectral sequence

$$E_1^{a,i-a} := H^{a,i-a}_{\mathrm{Hodge,c}}(Y, \mathcal{O}_Y) \Longrightarrow H^i_{\mathrm{dR,LLZ,c}}(Y)$$

on the E_1 page by [**?**, Theorem 3.1.10]. $\mathrm{tr}_{\mathrm{dR,LLZ}}$ extends to the derived level.

In [**?**], $\mathrm{tr}_{\mathrm{dR}}$ is defined for Y by the cosheaf condition: they construct $\mathrm{tr}_{\mathrm{dR}}$ for Stein spaces first, by using the coherent trace map for Stein spaces constructed in [**?**]. Note that the coherent trace map for Stein spaces in [**?**] is compatible with the one in [**?**]: it suffices to check it for an open disc, which follows from [**?**, Theorem 3.7] and a direct computation. In particular, we have

Lemma 4.20. *Via the natural quasi-isomorphism* $\mathrm{R}\Gamma_{\mathrm{dR,LLZ,c}}(Y) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{dR,c}}(Y)$, *we have*

$$\mathrm{tr}_{\mathrm{dR,LLZ}} = \mathrm{tr}_{\mathrm{dR}},$$

which is also Galois equivariant.

Proof. Since $\mathrm{tr}_{\mathrm{dR}}$ is compatible with open immersions, and $\mathrm{tr}_{\mathrm{dR,LLZ}}$ has similar compatibility by [**?**, Lemma 4.2.12], we may assume Y is proper. In this case, both are just the coherent trace map $\mathrm{tr}_{\mathrm{coh}} : H^d(Y, \Omega^d_{Y/K}) \to K$ via the canonical isomorphism $H^d(Y, \Omega^d_{Y/K}) \xrightarrow{\simeq} H^{2d}_{\mathrm{dR}}(Y)$.

To see that the de Rham trace map is Galois equivariant, we need to show the square

$$\begin{array}{ccc} H^d(Y_C, \Omega^d_{Y_C}) & \xrightarrow{\mathrm{tr}_{\mathrm{coh}}} & C \\ \downarrow & & \downarrow \sigma \\ H^d(Y_C^\sigma, \Omega^d_{Y_C^\sigma}) & \xrightarrow{\mathrm{tr}_{\mathrm{coh}}} & C. \end{array}$$

is commutative, where $\sigma \in \mathrm{Gal}(C/K)$. In fact, according to [**?**], we can take a classical point $y \in Y_C$ and the trace map $\mathrm{tr} : H^d_y(\Omega^d_{Y_C}) \to C$ is surjective and factors as

$$H^d_y(\Omega^d_{Y_C}) \to H^d(Y_C, \Omega^d_{Y_C}) \xrightarrow{\mathrm{tr}_{\mathrm{coh}}} C.$$

Since $\mathrm{tr}_{\mathrm{coh}}$ is an isomorphism, we are reduced to showing that the square

$$\begin{array}{ccc} H^d_y(\Omega^d_{Y_C}) & \xrightarrow{\mathrm{tr}} & C \\ \downarrow & & \downarrow \sigma \\ H^d_{y^\sigma}(\Omega^d_{Y_C^\sigma}) & \xrightarrow{\mathrm{tr}} & C \end{array}$$

is commutative, where y^σ is the preimage of y under $Y_C^\sigma \to Y_C$. This can be deduced from the construction of the residue map for local cohomology. □

In particular, we may safely use all propositions in [**?**] when Y is an almost proper rigid analytic variety over K.

Poincaré duality for Hyodo-Kato cohomology of open rigid analytic varieties is formulated as follows and can be deduced from Poincaré duality for de Rham cohomology.

Theorem 4.21. [**?**, Theorems 5.30 and 5.34] *Let Y be*

(1) a partially proper smooth rigid analytic variety or a quasi-compact smooth dagger variety over $L = C$ of dimension d, and denote $L_F = F^{\mathrm{nr}}$; or

(2) a proper smooth rigid analytic variety that is defined over $L = K$ and admits a proper semistable formal model over $\mathrm{Spf}(\mathcal{O}_K)$, and denote $L_F = F$.

Then:

(i) There is a natural trace map in the derived category of solid (ϕ, N)-modules over L_F:

$$\mathrm{tr}_{\mathrm{HK}} : \mathrm{R}\Gamma_{\mathrm{HK,c}}(Y)[2d] \to L_F\{-d\},$$

compatible with the Hyodo-Kato morphism.

(ii) The pairing

$$\mathrm{R}\Gamma_{\mathrm{HK}}(Y) \otimes^{L\blacksquare}_{L_F} \mathrm{R}\Gamma_{\mathrm{HK,c}}(Y)[2d] \to \mathrm{R}\Gamma_{\mathrm{HK,c}}(Y)[2d] \to L_F\{-d\}$$

is a perfect duality of solid (ϕ, N)-modules over L_F and induces the following isomorphisms:

$$H^i_{\mathrm{HK}}(Y) \simeq \mathrm{Hom}_{L_F}(H^{2d-i}_{\mathrm{HK,c}}(Y), L_F\{-d\}),$$

$$H^i_{\mathrm{HK,c}}(Y) \simeq \mathrm{Hom}_{L_F}(H^{2d-i}_{\mathrm{HK}}(Y), L_F\{-d\}).$$

Remark 4.22. As in Poincaré duality for de Rham cohomology, the trace map $\mathrm{tr}_{\mathrm{HK}}$ is compatible with restrictions to open dagger subvarieties.

Proof. See [**?**, Theorems 5.30 and 5.34]. The proof for case (2) is the same, since we have the Hyodo-Kato isomorphism in this case. □

The open-closed exact sequence, combined with Poincaré duality, gives us the exact Gysin sequence.

Proposition 4.23. *If X is a smooth dagger variety over K or C, and $Z \subset X$ is a smooth closed subvariety of pure codimension i which is nowhere dense, set $U := X - Z$. Then we have*

$$\mathrm{R}\Gamma_{\mathrm{dR}}(Z)[-2i] \simeq [\mathrm{R}\Gamma_{\mathrm{dR}}(X) \to \mathrm{R}\Gamma_{\mathrm{dR}}(U)],$$

$$\mathrm{R}\Gamma_{\mathrm{HK}}(Z)\{-i\}[-2i] \simeq [\mathrm{R}\Gamma_{\mathrm{HK}}(X) \to \mathrm{R}\Gamma_{\mathrm{HK}}(U)].$$

Proof. We prove the Gysin sequence for Hyodo-Kato cohomology; the proof for de Rham cohomology is the same (also proved in [**?**, Proposition 2.5]). We assume X is defined over C first. We may assume that X is quasi-compact. Applying Poincaré duality (Theorem **??**) to the open-closed exact sequence:

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(U) \simeq [\mathrm{R}\Gamma_{\mathrm{HK,c}}(X) \to \mathrm{R}\Gamma_{\mathrm{HK,c}}(Z)],$$

we get

$$\mathrm{R}\Gamma_{\mathrm{HK}}(Z)\{-i\}[-2i] \simeq [\mathrm{R}\Gamma_{\mathrm{HK}}(X) \to \mathrm{R}\Gamma_{\mathrm{HK}}(U)].$$

Remark **??** ensures the induced map $\mathrm{R}\Gamma_{\mathrm{HK}}(X) \to \mathrm{R}\Gamma_{\mathrm{HK}}(U)$ is exactly the natural map induced by the open immersion $U \hookrightarrow X$.

The case where X is defined over K then follows by Galois descent. □

Remark 4.24. The Gysin sequence is also true for étale cohomology, which follows from cohomological purity [**?**, 3.9].

5. Applications

In this section, we prove the comparison between algebraic and analytic Hyodo-Kato cohomology and the semistable conjecture for étale cohomology of almost proper rigid analytic varieties. We will also show that Tsuji's compactly supported log-crystalline cohomology (after inverting p) agrees with our definition of compactly supported Hyodo-Kato cohomology.

5.1. Algebraic and analytic Hyodo-Kato cohomology.

Recall that for an algebraic variety, we can compare the algebraic and analytic de Rham cohomology. This follows from [**?**, Theorem 2.3] and Theorem **??**.

Theorem 5.1. *Let X be an algebraic variety over L, and let X^{an} be its analytification. Then there exists a natural quasi-isomorphism*

$$\mathrm{R}\Gamma_{\mathrm{dR}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{dR}}(X^{\mathrm{an}}).$$

In this section, we will prove the same result for Hyodo-Kato cohomology. Note that for de Rham cohomology, we can construct the comparison map between algebraic and analytic de Rham cohomology directly from the projection $X_{\mathrm{an}} \to X_{\mathrm{zar}}$, and prove that it is an isomorphism using resolution of singularities and the existence of tubular neighborhoods after desingularization. For Hyodo-Kato cohomology, it is unclear how to define the comparison map. On the other hand, once we construct a map $\mathrm{R}\Gamma_{\mathrm{HK}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(X^{\mathrm{an}})$ that is compatible, via the Hyodo-Kato isomorphism, with its de Rham counterpart, the naturality of this map follows from that of its de Rham counterpart. If X is defined over K, we can base change to the geometric setting and then use Galois descent.

Theorem 5.2. *Let X be an algebraic variety over K, and let X^{an} be its analytification. Then there exists a natural quasi-isomorphism*

$$\mathrm{R}\Gamma_{\mathrm{HK}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(X^{\mathrm{an}}),$$

which is compatible with Frobenius, monodromy, and the GAGA morphism for de Rham cohomology, i.e., we have the following commutative square:

$$\begin{array}{ccc} \mathrm{R}\Gamma_{\mathrm{HK}}(X) & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{dR}}(X) \\ \downarrow\simeq & & \downarrow\simeq \\ \mathrm{R}\Gamma_{\mathrm{HK}}(X^{\mathrm{an}}) & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{dR}}(X^{\mathrm{an}}), \end{array}$$

where the horizontal maps are Hyodo-Kato morphisms.

In the geometric case, we also need to base change along $\overline{K} \to C$.

Theorem 5.3. *Let X be an algebraic variety over $\overline{K}$, and let X_C^{an} be the analytification of $X_C := X \times_{\overline{K}} C$. Then there exists a natural quasi-isomorphism*

$$\mathrm{R}\Gamma_{\mathrm{HK}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(X_C^{\mathrm{an}}),$$

which is compatible with the Galois action (when X is the base change of some X_0 over K), Frobenius, monodromy, and the GAGA morphism for de Rham cohomology.

5.1.1. *The local case.* We first prove the result for the Beilinson basis, i.e., the following proposition.

Proposition 5.4. *Suppose $(U, \overline{U})$ is a strictly semistable pair over K, as in the discussion preceding Proposition* **??**. *Then we have quasi-isomorphisms, natural with respect to all strictly semistable pairs,*

$$\mathrm{R}\Gamma_{\mathrm{HK}}(U) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(U^{\mathrm{an}}),$$
$$\mathrm{R}\Gamma_{\mathrm{HK}}(U_{\overline{K}}) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(U_C^{\mathrm{an}}).$$

Moreover, both are compatible with Frobenius, monodromy, and the GAGA morphism for de Rham cohomology, and the second is compatible with the Galois action.

Proof. Consider a strictly semistable pair $(U, \overline{U})$. Let $Z = \overline{U}_K - U$. Local-global compatibility for algebraic varieties (Proposition **??**) and rigid analytic varieties (Proposition **??**) implies the following lemma.

Lemma 5.5. *We have natural quasi-isomorphisms*

$$\mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_K) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_K^{\mathrm{an}}), \quad \mathrm{R}\Gamma_{\mathrm{HK}}(Z) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(Z^{\mathrm{an}}),$$

which are compatible with Frobenius, monodromy, and the GAGA morphism for de Rham cohomology.

Proof. Since the p-adic completion of $\overline{U}$ is a semistable formal model of $\overline{U}_K^{\mathrm{an}}$, the first quasi-isomorphism can be defined as the identity.

For the second isomorphism, if Z is smooth, Lemma **??** shows that the p-adic completion of its closure $\overline{Z}$ in $\overline{U}$ is a semistable formal model of Z^{an}. We therefore have the natural quasi-isomorphism.

In general, write $Z = D_1 \cup \cdots \cup D_n$, where the D_i are the irreducible components of Z. By h-descent for algebraic varieties and éh-descent for rigid analytic varieties, we can reduce to the smooth intersections of the D_i.

To show compatibility with the GAGA morphism for de Rham cohomology, note that for $\overline{U}$ we have the following diagram:

$$\begin{array}{ccccc}
\mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_1^0) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_K) & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{dR}}(\overline{U}_K) \\
\| & & \downarrow\simeq & & \downarrow\simeq \\
\mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_1^0) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_K^{\mathrm{an}}) & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{dR}}(\overline{U}_K^{\mathrm{an}}),
\end{array}$$

where the left square is commutative. To show that the right square is commutative, it suffices to prove that the outer rectangle is commutative. The horizontal composites are given by

$$\begin{array}{ccccccc}
\mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_1^0) & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_1^0)\otimes_F K & \xrightarrow{\iota_{\mathrm{HK}}} & \mathrm{R}\Gamma_{\mathrm{cris}}(\overline{U}_1/O_K^\times)_{\mathbb{Q}_p} & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR}}(\overline{U}_K) \\
\| & & \| & & \| & & \downarrow\simeq \\
\mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_1^0) & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_1^0)\otimes_F K & \xrightarrow{\iota_{\mathrm{HK}}} & \mathrm{R}\Gamma_{\mathrm{cris}}(\overline{U}_1/O_K^\times)_{\mathbb{Q}_p} & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR}}(\overline{U}_K^{\mathrm{an}}),
\end{array}$$

and we are reduced to showing that the rightmost square is commutative. In fact, we can prove an integral version of this commutativity, i.e., that the diagram

$$\begin{array}{ccc}
 & & \mathrm{R}\Gamma_{\mathrm{dR}}(\overline{U}/O_K^\times) \\
\mathrm{R}\Gamma_{\mathrm{cris}}(\overline{U}_1/O_K^\times) = \lim_n \mathrm{R}\Gamma_{\mathrm{cris}}(\overline{U}_1/O_{K,n}^\times) \simeq \lim_n \mathrm{R}\Gamma_{\mathrm{dR}}(\overline{U}_1/O_{K,n}^\times) & & \downarrow \\
 & & \mathrm{R}\Gamma_{\mathrm{dR}}(\overline{\mathcal{U}}/O_K^\times)
\end{array}$$

is commutative, where $\overline{\mathcal{U}}$ is the p-adic completion of $\overline{U}$. Here we use the fact that the rigid generic fiber of $\overline{\mathcal{U}}$ coincides with $\overline{U}_K^{\mathrm{an}}$. This commutativity follows from the construction in the theorem on formal functions, as described in [**?**, Section 4].

By reducing to the smooth case, the same argument also shows the compatibility for Z. □

By Definition **??** of compactly supported Hyodo-Kato cohomology, we have

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(U) \simeq [\mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_K) \to \mathrm{R}\Gamma_{\mathrm{HK}}(Z)].$$

By Proposition **??**, we also have

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(U^{\mathrm{an}}) \simeq [\mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_K^{\mathrm{an}}) \to \mathrm{R}\Gamma_{\mathrm{HK}}(Z^{\mathrm{an}})].$$

Therefore, the above lemma shows that we have the natural quasi-isomorphism

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(U) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK,c}}(U^{\mathrm{an}}),$$

and the same argument gives $\mathrm{R}\Gamma_{\mathrm{HK,c}}(U_{\overline{K}}) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK,c}}(U_C^{\mathrm{an}})$, compatible with the Hyodo-Kato morphism for compactly supported cohomology. Since U is smooth, Poincaré duality gives the quasi-isomorphism $\mathrm{R}\Gamma_{\mathrm{HK}}(U_{\overline{K}}) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(U_C^{\mathrm{an}})$ as the dotted arrow in the following diagram:

$$\begin{array}{ccc} \mathrm{R}\Gamma_{\mathrm{HK}}(U_{\overline{K}}) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK,c}}(U_{\overline{K}})^\vee \\ \vdots & & \uparrow\simeq \\ \mathrm{R}\Gamma_{\mathrm{HK}}(U_C^{\mathrm{an}}) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK,c}}(U_C^{\mathrm{an}})^\vee. \end{array}$$

Here the right vertical map is the dual of the comparison map for compactly supported cohomology. Both the algebraic and analytic Hyodo-Kato trace maps are compatible with the corresponding de Rham trace maps under the Hyodo-Kato morphism, and the algebraic and analytic de Rham trace maps are compatible with the GAGA morphism ([**?**, Lemma 5.4]).

This also implies that the isomorphism $\mathrm{R}\Gamma_{\mathrm{HK}}(U_{\overline{K}}) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(U_C^{\mathrm{an}})$ is compatible with the Hyodo-Kato morphism, as we have the following commutative diagram:

$$\begin{array}{ccccccc} \mathrm{R}\Gamma_{\mathrm{HK}}(U_{\overline{K}}) \otimes_{F^{\mathrm{nr}}} C & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK,c}}(U_{\overline{K}})^\vee \otimes_{F^{\mathrm{nr}}} C & \xleftarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR,c}}(U_C)^\vee & \xleftarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR}}(U_C) \\ \downarrow\simeq & & \uparrow\simeq & & \uparrow\simeq & & \downarrow\simeq \\ \mathrm{R}\Gamma_{\mathrm{HK}}(U_C^{\mathrm{an}}) \otimes_{F^{\mathrm{nr}}} C & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK,c}}(U_C^{\mathrm{an}})^\vee \otimes_{F^{\mathrm{nr}}} C & \xleftarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR,c}}(U_C^{\mathrm{an}})^\vee & \xleftarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR}}(U_C^{\mathrm{an}}). \end{array}$$

Since Hyodo-Kato cohomology satisfies Galois descent, taking Galois invariants gives

$$\mathrm{R}\Gamma_{\mathrm{HK}}(U) \simeq \mathrm{R}\Gamma_{\mathrm{HK}}(U^{\mathrm{an}}),$$

which is also compatible with the Hyodo-Kato morphism.

Finally, our construction is functorial, i.e., for a morphism of semistable pairs $(U', \overline{U'}) \to (U, \overline{U})$, where $(U, \overline{U})$ is a semistable pair over K and $(U', \overline{U'})$ is a semistable pair over K', we have a commutative square

$$\begin{array}{ccc} \mathrm{R}\Gamma_{\mathrm{HK}}(U) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK}}(U^{\mathrm{an}}) \\ \downarrow & & \downarrow \\ \mathrm{R}\Gamma_{\mathrm{HK}}(U') & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK}}(U'^{\mathrm{an}}), \end{array}$$

and a similar commutative square over $\overline{K}$. To see this, we may assume that K' is a finite extension of K with residue field k', and let F' be the fraction field of $W(k')$. After tensoring the upper row

with K' over F and the lower row with K' over F', the Hyodo-Kato isomorphism reduces the claim to the commutativity of the square

$$\begin{array}{ccc} \mathrm{R}\Gamma_{\mathrm{dR}}(U_{K'}) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR}}(U_{K'}^{\mathrm{an}}) \\ \downarrow & & \downarrow \\ \mathrm{R}\Gamma_{\mathrm{dR}}(U') & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR}}(U'^{\mathrm{an}}), \end{array}$$

which follows from the functoriality of the de Rham GAGA morphism. □

5.1.2. *The global case.* Let X be a general algebraic variety over $\overline{K}$, and take an h-hypercover $(U^\bullet, \overline{U}^\bullet) \to X$. Since we have proved GAGA for local models, we have

$$\begin{array}{ccc} \mathrm{R}\Gamma_{\mathrm{HK}}(X) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK}}(U^\bullet) \\ \vdots & & \downarrow\simeq \\ \mathrm{R}\Gamma_{\mathrm{HK}}(X_C^{\mathrm{an}}) & \xrightarrow{\psi} & \mathrm{R}\Gamma_{\mathrm{HK}}(U_C^{\mathrm{an},\bullet}). \end{array}$$

To construct the dotted arrow, it suffices to show that the bottom arrow is a quasi-isomorphism. In general, we do not know whether h-descent holds for rigid analytic varieties. However, we can use the Hyodo-Kato morphism and GAGA for de Rham cohomology. Tensoring the solid arrows with C over F^{nr} and using the Hyodo-Kato isomorphisms identifies them with three sides of the commutative diagram

$$\begin{array}{ccc} \mathrm{R}\Gamma_{\mathrm{dR}}(X_C) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR}}(U_C^\bullet) \\ \downarrow\simeq & & \downarrow\simeq \\ \mathrm{R}\Gamma_{\mathrm{dR}}(X_C^{\mathrm{an}}) & \xrightarrow{\phi} & \mathrm{R}\Gamma_{\mathrm{dR}}(U_C^{\mathrm{an},\bullet}), \end{array}$$

where the vertical maps are the GAGA morphisms for de Rham cohomology. Thus ψ is a quasi-isomorphism, since $\phi = \psi \otimes_{F^{\mathrm{nr}}} C$ is a quasi-isomorphism. We can therefore construct a natural quasi-isomorphism $\mathrm{R}\Gamma_{\mathrm{HK}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(X_C^{\mathrm{an}})$. By construction and Proposition **??**, this quasi-isomorphism is compatible with the Galois action when X descends to K, as well as with Frobenius, monodromy, and the GAGA morphism for de Rham cohomology.

When X is defined over K, we can use the same construction: with the same notation, ψ is still a quasi-isomorphism by Galois descent.

5.1.3. *Algebraic and analytic cohomology with compact support.* We now prove GAGA for de Rham and Hyodo-Kato cohomology with compact support.

Theorem 5.6. *Let X be an algebraic variety over C, and let X^{an} be its analytification. Then we have a natural quasi-isomorphism*

$$\mathrm{R}\Gamma_{\mathrm{dR,c}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{dR,c}}(X^{\mathrm{an}}),$$

which is compatible with the Galois action when X descends to K. Moreover, the morphism makes the following square commute:

$$\begin{array}{ccc} \mathrm{R}\Gamma_{\mathrm{dR,c}}(X) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR,c}}(X^{\mathrm{an}}) \\ \downarrow & & \downarrow \\ \mathrm{R}\Gamma_{\mathrm{dR}}(X) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR}}(X^{\mathrm{an}}). \end{array} \tag{5.1}$$

Proof. When X is smooth, the theorem follows from GAGA for ordinary cohomology and Poincaré duality, i.e., $\mathrm{R}\Gamma_{\mathrm{dR,c}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{dR,c}}(X^{\mathrm{an}})$ is defined as the dotted arrow in the following commutative square:

$$\begin{array}{ccc} \mathrm{R}\Gamma_{\mathrm{dR,c}}(X) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR}}(X)^{\vee} \\ \vdots & & \simeq\uparrow \\ \mathrm{R}\Gamma_{\mathrm{dR,c}}(X^{\mathrm{an}}) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{dR}}(X^{\mathrm{an}})^{\vee}. \end{array}$$

To see that the square (**??**) is commutative, choose a smooth compactification $\overline{X}$ such that $\overline{X} - X$ is a strict normal crossings divisor, using Hironaka's resolution of singularities ([**?**]). Then the commutativity of (**??**) follows directly from GAGA for coherent sheaves ([**?**]) and the description of compactly supported de Rham cohomology in [**?**, Definition 3.1.1].

In general, we use induction on the dimension of X. When $\dim X = 0$, the theorem is clear. By Hironaka's resolution of singularities ([**?**]), there exists a finite sequence of blowups along smooth centers $X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = X$ such that X_n is smooth. We now show that the theorem holds for X_{n-1}. Let $Y_{n-1} \hookrightarrow X_{n-1}$ be the blowup center, and let $Y_n \hookrightarrow X_n$ be its inverse image. Then Proposition **??** gives

$$\mathrm{R}\Gamma_{\mathrm{dR,c}}(X_{n-1}) \simeq [\mathrm{R}\Gamma_{\mathrm{dR,c}}(Y_{n-1}) \oplus \mathrm{R}\Gamma_{\mathrm{dR,c}}(X_n) \to \mathrm{R}\Gamma_{\mathrm{dR,c}}(Y_n)].$$

By the induction hypothesis and the smoothness of X_n, the theorem holds for Y_n, Y_{n-1}, and X_n. A diagram chase therefore shows that it also holds for X_{n-1}. Repeating this argument for $X_{n-2}, \ldots, X_0$ concludes the proof. □

Theorem 5.7. *Let X be an algebraic variety over $\overline{K}$, and let X_C^{an} be the analytification of $X_C := X \times_{\overline{K}} C$. Then we have a natural quasi-isomorphism*

$$\mathrm{R}\Gamma_{\mathrm{HK,c}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK,c}}(X_C^{\mathrm{an}}),$$

which is compatible with Frobenius, monodromy, the GAGA morphism for de Rham cohomology with compact support, and the Galois action when X descends to K. Moreover, the morphism makes the following square commute:

$$\begin{array}{ccc} \mathrm{R}\Gamma_{\mathrm{HK,c}}(X) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK,c}}(X_C^{\mathrm{an}}) \\ \downarrow & & \downarrow \\ \mathrm{R}\Gamma_{\mathrm{HK}}(X) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK}}(X_C^{\mathrm{an}}). \end{array} \tag{5.2}$$

Proof. The proof is the same as that of compactly supported de Rham GAGA. The square (**??**) is commutative because it becomes commutative after tensoring with C over F^{nr}. □

Remark 5.8. Our constructions are compatible with Poincaré duality. Namely, for a smooth algebraic variety X of pure dimension d over K, they fit into the commutative diagram

$$\begin{array}{ccccccc} \mathrm{R}\Gamma_{\mathrm{HK}}(X) & \otimes_F^{L\blacksquare} & \mathrm{R}\Gamma_{\mathrm{HK,c}}(X)[2d] & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{HK,c}}(X)[2d] & \xrightarrow{\mathrm{tr}} & F\{-d\} \\ \downarrow\simeq & & \downarrow\simeq & & \downarrow\simeq & & \| \\ \mathrm{R}\Gamma_{\mathrm{HK}}(X^{\mathrm{an}}) & \otimes_F^{L\blacksquare} & \mathrm{R}\Gamma_{\mathrm{HK,c}}(X^{\mathrm{an}})[2d] & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{HK,c}}(X^{\mathrm{an}})[2d] & \xrightarrow{\mathrm{tr}} & F\{-d\}. \end{array}$$

The analogous diagram holds over $\overline{K}$, with F replaced by F^{nr} and X^{an} replaced by X^{an}_C.

5.2. **Semistable conjecture for open varieties.** In this section, we prove the following theorem, which extends the semistable conjecture from proper smooth rigid analytic varieties to almost proper smooth rigid analytic varieties.

Theorem 5.9. *Suppose X is a proper smooth rigid analytic variety over C, $Z \subset X$ is a strict normal crossings divisor, and $U = X - Z$.*

(1) We have a B_{st}-linear functorial isomorphism commuting with φ and N

$$\alpha^i_{\mathrm{st}}(U) : H^i_{\text{ét}}(U, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \simeq H^i_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}},$$

which induces a B_{dR}-linear filtered isomorphism

$$H^i_{\text{ét}}(U, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{dR}} \simeq H^i_{B^+_{\mathrm{dR}}}(X) \otimes_{B^+_{\mathrm{dR}}} B_{\mathrm{dR}}.$$

Here, the filtration on $H^i_{B^+_{\mathrm{dR}}}(X)$ is defined by

$$\mathrm{Fil}^\star H^i_{B^+_{\mathrm{dR}}}(X) := \mathrm{Im}(H^i(\mathrm{Fil}^\star \mathrm{R}\Gamma_{B^+_{\mathrm{dR}}}(X)) \to H^i_{B^+_{\mathrm{dR}}}(X)).$$

(2) Let $i \leq r$. Then we have an exact sequence

$$0 \to H^i_{\text{ét}}(U, \mathbb{Q}_p(r)) \to (H^i_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B^+_{\mathrm{st}})^{N=0, \varphi=p^r} \to H^i_{B^+_{\mathrm{dR}}}(X)/\mathrm{Fil}^r \to 0.$$

Moreover, when X descends to a rigid analytic variety over K, the statements in (1) and (2) are Galois equivariant.

Proof. We prove the theorem in the following steps. Let $d = \dim X$.

(1) Construction of the comparison map. We can define the period morphism

$$\alpha_{\mathrm{st}}(U) : \mathrm{R}\Gamma_{\text{ét}}(U, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \to \mathrm{R}\Gamma_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}},$$

as follows. For $r > 2d$, consider the following composition

$$\begin{aligned} \mathrm{R}\Gamma_{\text{ét}}(U, \mathbb{Q}_p(r)) &\to \tau^{\leq 2d} \mathrm{R}\Gamma_{\text{proét}}(U, \mathbb{Q}_p(r)) \xrightarrow{\simeq} \tau^{\leq 2d} \mathrm{R}\Gamma_{\mathrm{syn}}(U, r) \\ &\to [\mathrm{R}\Gamma_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}]^{N=0, \varphi=p^r} \\ &\xrightarrow{p^{-r}} \mathrm{R}\Gamma_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}. \end{aligned}$$

Denote this composition by $\alpha_{\mathrm{st}}(r)$. We set

$$\alpha_{\mathrm{st}}(U) := t^{-r}\alpha_{\mathrm{st}}(r)\varepsilon^r,$$

where ε is the generator of $\mathbb{Z}_p(1)$ corresponding to t.

Note that α_{st} is natural with respect to any morphism $V \to U$, since each map in the construction of α_{st} has this property.

(2) Compatibility with the Gysin morphism. Suppose that Z is smooth. We check that the diagram

$$\begin{array}{ccccc} \mathrm{R}\Gamma_{\text{ét}}(Z,\mathbb{Q}_p(-1))[-2]\otimes_{\mathbb{Q}_p} B_{\mathrm{st}} & \xrightarrow{g_{\text{ét}}} & \mathrm{R}\Gamma_{\text{ét}}(X,\mathbb{Q}_p)\otimes_{\mathbb{Q}_p} B_{\mathrm{st}} & \longrightarrow & \mathrm{R}\Gamma_{\text{ét}}(U,\mathbb{Q}_p)\otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \\ \downarrow{\scriptstyle \alpha_{\mathrm{st}}(Z)(-1)[-2]} & & \downarrow{\scriptstyle \alpha_{\mathrm{st}}(X)} & & \downarrow{\scriptstyle \alpha_{\mathrm{st}}(U)} \\ \mathrm{R}\Gamma_{\mathrm{HK}}(Z)\{-1\}[-2]\otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}} & \xrightarrow{g_{\mathrm{HK}}} & \mathrm{R}\Gamma_{\mathrm{HK}}(X)\otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}} & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{HK}}(U)\otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}, \end{array} \tag{5.3}$$

commutes. The right square is commutative, as α_{st} is natural with respect to the open immersion $U \to X$. For the left square, by Theorem **??**, we know that $\alpha_{\mathrm{st}}(Z)$ and $\alpha_{\mathrm{st}}(X)$ are quasi-isomorphisms.

Denote by $PD_{\text{ét}}(X)$ and $PD_{\mathrm{HK}}(X)$ the Poincaré duality morphisms

$$PD_{\text{ét}}(X) : \mathrm{R}\Gamma_{\text{ét}}(X,\mathbb{Q}_p)\otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \to R\,\mathrm{Hom}_{B_{\mathrm{st}}}(\mathrm{R}\Gamma_{\text{ét}}(X,\mathbb{Q}_p(d))[2d]\otimes_{\mathbb{Q}_p} B_{\mathrm{st}}, B_{\mathrm{st}}),$$

$$PD_{\mathrm{HK}}(X) : \mathrm{R}\Gamma_{\mathrm{HK}}(X)\otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}} \to R\,\mathrm{Hom}_{B_{\mathrm{st}}}(\mathrm{R}\Gamma_{\mathrm{HK}}(X)\{d\}[2d]\otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}, B_{\mathrm{st}}).$$

Here the trace map for étale cohomology is just the trace map for pro-étale cohomology: this indeed gives a perfect pairing, for example, by the main theorem of [**?**]. $PD_{\text{ét}}(X)$ is a quasi-isomorphism, as for any i,

$$H^i(\mathrm{R}\Gamma_{\text{ét}}(X,\mathbb{Q}_p(d))[2d]\otimes_{\mathbb{Q}_p} B_{\mathrm{st}}) \simeq H^{i+2d}_{\text{ét}}(X,\mathbb{Q}_p(d))\otimes_{\mathbb{Q}_p} B_{\mathrm{st}}$$

is a finite free B_{st}-module. Therefore, for any $j > 0$,

$$\mathrm{Ext}^j_{B_{\mathrm{st}}}(H^{i+2d}_{\text{ét}}(X,\mathbb{Q}_p(d))\otimes_{\mathbb{Q}_p} B_{\mathrm{st}}, B_{\mathrm{st}}) = 0.$$

The same argument also shows that $PD_{\mathrm{HK}}(X)$ is a quasi-isomorphism. Define $PD_{\text{ét}}(Z)$ and $PD_{\mathrm{HK}}(Z)$ similarly. Let $i_{\text{ét}}$ and i_{HK} be the canonical maps

$$i_{\text{ét}} : \mathrm{R}\Gamma_{\text{ét}}(X,\mathbb{Q}_p)\otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \to \mathrm{R}\Gamma_{\text{ét}}(Z,\mathbb{Q}_p)\otimes_{\mathbb{Q}_p} B_{\mathrm{st}},$$

$$i_{\mathrm{HK}} : \mathrm{R}\Gamma_{\mathrm{HK}}(X)\otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}} \to \mathrm{R}\Gamma_{\mathrm{HK}}(Z)\otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}.$$

Then for $*$ =HK or ét, the Gysin morphism g_* is

$$g_* := PD_*(X)^{-1}\circ R\,\mathrm{Hom}_{B_{\mathrm{st}}}(i_*[2d], B_{\mathrm{st}})\circ PD_*(Z)[-2].$$

We need to show that

$$\alpha_{\mathrm{st}}(X)\circ g_{\text{ét}} = g_{\mathrm{HK}}\circ\alpha_{\mathrm{st}}(Z)(-1)[-2].$$

We have the following lemma.

Lemma 5.10. *(1)* $\alpha_{\mathrm{st}}(Z)\circ i_{\text{ét}} = i_{\mathrm{HK}}\circ\alpha_{\mathrm{st}}(X)$.

(2) $R\,\mathrm{Hom}_{B_{\mathrm{st}}}(\alpha_{\mathrm{st}}(X)(d)[2d], B_{\mathrm{st}})\circ PD_{\mathrm{HK}}(X)\circ\alpha_{\mathrm{st}}(X) = PD_{\text{ét}}(X)$.

(3) $R\,\mathrm{Hom}_{B_{\mathrm{st}}}(\alpha_{\mathrm{st}}(Z)(d-1)[2d-2], B_{\mathrm{st}})\circ PD_{\mathrm{HK}}(Z)\circ\alpha_{\mathrm{st}}(Z) = PD_{\text{ét}}(Z)$.

Proof. The first assertion follows from the fact that α_{st} is natural with respect to the morphism $Z \to X$.

We prove assertion (2) for X; the proof of (3) for Z is the same. Recall the construction of α_{st}. Choose $r, r' > 2d$ and set $s := r + r' - d$. After twisting, we need to show that the following diagram

$$\begin{array}{ccccccc}
\mathrm{R}\Gamma_{\text{ét}}(X,\mathbb{Q}_p(r)) & \otimes^{L}_{\mathbb{Q}_p} & \mathrm{R}\Gamma_{\text{ét}}(X,\mathbb{Q}_p(r'))[2d] & \longrightarrow & \mathrm{R}\Gamma_{\text{ét}}(X,\mathbb{Q}_p(r+r'))[2d] & \xrightarrow{\mathrm{tr}_{\text{ét}}} & \mathbb{Q}_p(s) \\
\downarrow & & \downarrow & & \downarrow & & \| \\
\mathrm{R}\Gamma_{\text{proét}}(X,\mathbb{Q}_p(r)) & \otimes^{L}_{\mathbb{Q}_p} & \mathrm{R}\Gamma_{\text{proét}}(X,\mathbb{Q}_p(r'))[2d] & \longrightarrow & \mathrm{R}\Gamma_{\text{proét}}(X,\mathbb{Q}_p(r+r'))[2d] & \xrightarrow{\mathrm{tr}_{\text{proét}}} & \mathbb{Q}_p(s) \\
{\scriptstyle\alpha_r}\uparrow & & {\scriptstyle\alpha_{r'}}\uparrow & & {\scriptstyle\alpha_{r'+r}}\uparrow & & \| \\
\mathrm{R}\Gamma_{\mathrm{syn}}(X,r) & \otimes^{L}_{\mathbb{Q}_p} & \mathrm{R}\Gamma_{\mathrm{syn}}(X,r')[2d] & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{syn}}(X,r+r')[2d] & \xrightarrow{\mathrm{tr}_{\mathrm{syn}}} & \mathbb{Q}_p(s) \\
\downarrow & & \downarrow & & \downarrow & & \downarrow \\
\mathrm{R}\Gamma_{\mathrm{HK}}(X)\{r\}\otimes B_{\mathrm{st}} & \otimes^{L}_{B_{\mathrm{st}}} & \mathrm{R}\Gamma_{\mathrm{HK}}(X)\{r'\}\otimes B_{\mathrm{st}}[2d] & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{HK}}(X)\{r+r'\}\otimes B_{\mathrm{st}}[2d] & \xrightarrow{\mathrm{tr}^{\mathrm{HK}}_{B_{\mathrm{st}}}} & B_{\mathrm{st}}\{s\}.
\end{array}$$

commutes after applying the truncation $\tau^{\leq 2d}$. Here the α's are period morphisms from [**?**, Theorem 6.9].

We only need to consider the pairings. The top two rows are naturally compatible. The bottom two rows are compatible because the product on syntomic cohomology is defined using the mapping fiber. The middle two rows are compatible because the classical comparison maps are compatible with products. □

Now we can calculate

$$\begin{aligned}
\alpha_{\mathrm{st}}(X)\circ g_{\text{ét}} &= \alpha_{\mathrm{st}}(X)\circ PD_{\text{ét}}(X)^{-1}\circ R\operatorname{Hom}_{B_{\mathrm{st}}}(i_{\text{ét}}[2d], B_{\mathrm{st}})\circ PD_{\text{ét}}(Z)[-2] \\
&= PD^{-1}_{\mathrm{HK}}(X)\circ R\operatorname{Hom}_{B_{\mathrm{st}}}(\alpha^{-1}_{\mathrm{st}}(X)(d)[2d], B_{\mathrm{st}})\circ R\operatorname{Hom}_{B_{\mathrm{st}}}(i_{\text{ét}}[2d], B_{\mathrm{st}})\circ PD_{\text{ét}}(Z)[-2] \\
&= PD^{-1}_{\mathrm{HK}}(X)\circ R\operatorname{Hom}_{B_{\mathrm{st}}}(i_{\text{ét}}[2d]\circ\alpha^{-1}_{\mathrm{st}}(X)(d)[2d], B_{\mathrm{st}})\circ PD_{\text{ét}}(Z)[-2] \\
&= PD^{-1}_{\mathrm{HK}}(X)\circ R\operatorname{Hom}_{B_{\mathrm{st}}}(\alpha^{-1}_{\mathrm{st}}(Z)(d)[2d]\circ i_{\mathrm{HK}}[2d], B_{\mathrm{st}})\circ PD_{\text{ét}}(Z)[-2] \\
&= PD^{-1}_{\mathrm{HK}}(X)\circ R\operatorname{Hom}_{B_{\mathrm{st}}}(i_{\mathrm{HK}}[2d], B_{\mathrm{st}})\circ R\operatorname{Hom}_{B_{\mathrm{st}}}(\alpha^{-1}_{\mathrm{st}}(Z)(d)[2d], B_{\mathrm{st}})\circ PD_{\text{ét}}(Z)[-2] \\
&= PD^{-1}_{\mathrm{HK}}(X)\circ R\operatorname{Hom}_{B_{\mathrm{st}}}(i_{\mathrm{HK}}[2d], B_{\mathrm{st}})\circ PD_{\mathrm{HK}}(Z)[-2]\circ\alpha_{\mathrm{st}}(Z)(-1)[-2] \\
&= g_{\mathrm{HK}}\circ\alpha_{\mathrm{st}}(Z)(-1)[-2].
\end{aligned}$$

This proves compatibility with the Gysin morphism.

(3) α_{st} is a natural quasi-isomorphism compatible with Gysin morphisms. We assume first that Z is smooth. From (2), we know that $\alpha_{\mathrm{st}}(Z)$ and $\alpha_{\mathrm{st}}(X)$ are isomorphisms compatible with Gysin morphisms. Therefore the commutative diagram in step (2) shows that we have a natural quasi-isomorphism:

$$\alpha_{\mathrm{st}}(U) : \mathrm{R}\Gamma_{\text{ét}}(U,\mathbb{Q}_p)\otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(U)\otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}.$$

In general, we can write Z as the union of irreducible components $Z = Z_1 \cup \cdots \cup Z_m$, where $Z_1, \ldots, Z_m$ are smooth. Write $D_j := \bigcup_{i \leq j} Z_i$. We have a map of distinguished triangles:

$$\begin{array}{ccccc}
\mathrm{R}\Gamma_{\text{ét}}(Z_m - Z_m \cap D_{m-1}, \mathbb{Q}_p(-1))[-2] \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} & \longrightarrow & \mathrm{R}\Gamma_{\text{ét}}(X - D_{m-1}, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} & \longrightarrow & \mathrm{R}\Gamma_{\text{ét}}(U, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \\
\downarrow{\scriptstyle \alpha_{\mathrm{st}}(Z_m - Z_m \cap D_{m-1})(-1)[-2]} & & \downarrow{\scriptstyle \alpha_{\mathrm{st}}(X - D_{m-1})} & & \downarrow{\scriptstyle \alpha_{\mathrm{st}}(U)} \\
\mathrm{R}\Gamma_{\mathrm{HK}}(Z_m - Z_m \cap D_{m-1})\{-1\}[-2] \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}} & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{HK}}(X - D_{m-1}) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}} & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}},
\end{array}$$

We need to check that the left square is commutative. We use induction on the number of irreducible divisors m. Suppose that the left square is commutative for any proper smooth rigid analytic variety over C whose boundary has at most $m-1$ irreducible components. Since the Gysin map is functorial, we have the commutative diagram

$$\begin{array}{ccc}
\mathrm{R}\Gamma_{\text{ét}}((Z_m \cap Z_{m-1}) \setminus D_{m-2}, \mathbb{Q}_p(-2))[-4] & \longrightarrow & \mathrm{R}\Gamma_{\text{ét}}(Z_{m-1} - Z_{m-1} \cap D_{m-2}, \mathbb{Q}_p(-1))[-2] \\
\downarrow & & \downarrow \\
\mathrm{R}\Gamma_{\text{ét}}(Z_m - Z_m \cap D_{m-2}, \mathbb{Q}_p(-1))[-2] & \longrightarrow & \mathrm{R}\Gamma_{\text{ét}}(X - D_{m-2}, \mathbb{Q}_p) \\
\downarrow & & \downarrow \\
\mathrm{R}\Gamma_{\text{ét}}(Z_m - Z_m \cap D_{m-1}, \mathbb{Q}_p(-1))[-2] & \longrightarrow & \mathrm{R}\Gamma_{\text{ét}}(X - D_{m-1}, \mathbb{Q}_p),
\end{array}$$

where the horizontal arrows (and the first vertical arrow) are Gysin maps. We also have a similar commutative diagram for Hyodo-Kato cohomology. Therefore it suffices to show that the diagrams

$$\begin{array}{ccc}
\mathrm{R}\Gamma_{\text{ét}}(Z_{m-1} - Z_{m-1} \cap D_{m-2}, \mathbb{Q}_p(-1))[-2] \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} & \longrightarrow & \mathrm{R}\Gamma_{\text{ét}}(X - D_{m-2}, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \\
\downarrow & & \downarrow \\
\mathrm{R}\Gamma_{\mathrm{HK}}(Z_{m-1} - Z_{m-1} \cap D_{m-2})\{-1\}[-2] \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}} & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{HK}}(X - D_{m-2}) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}
\end{array}$$

and

$$\begin{array}{ccc}
\mathrm{R}\Gamma_{\text{ét}}((Z_m \cap Z_{m-1}) \setminus D_{m-2}, \mathbb{Q}_p(-1))[-2] \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} & \longrightarrow & \mathrm{R}\Gamma_{\text{ét}}(Z_{m-1} - Z_{m-1} \cap D_{m-2}, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \\
\downarrow & & \downarrow \\
\mathrm{R}\Gamma_{\mathrm{HK}}((Z_m \cap Z_{m-1}) \setminus D_{m-2})\{-1\}[-2] \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}} & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{HK}}(Z_{m-1} - Z_{m-1} \cap D_{m-2}) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}
\end{array}$$

are commutative. In both cases, the number of irreducible components of the boundary is at most $m-1$, so the induction hypothesis applies. Therefore we have a natural quasi-isomorphism:

$$\alpha_{\mathrm{st}}(U) : \mathrm{R}\Gamma_{\text{ét}}(U, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{st}} \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}}.$$

(4) The short exact sequence. The following theorem (see [**?**, Remark 5.16]) is an extension of the standard results for admissible filtered (φ, N)-modules, e.g., [**?**, Proposition 5.3] or [**?**, Chapter 10].

Theorem 5.11. *Suppose that (D, D^+_{dR}) is an acyclic filtered (φ, N)-module over C (see* [**?**, Definition 5.3.1] *for the definition) with φ-slopes in $[0, r]$, and $D \otimes t^r B^+_{\mathrm{dR}} \subset t^r D^+_{\mathrm{dR}} \subset D \otimes B^+_{\mathrm{dR}}$. Then we have the following short exact sequence:*

$$0 \to t^r V_{\mathrm{st}}(D, D^+_{\mathrm{dR}}) \to (D \otimes_{F^{\mathrm{nr}}} B^+_{\mathrm{st}})^{N=0, \varphi=p^r} \to (D \otimes_{F^{\mathrm{nr}}} B^+_{\mathrm{dR}})/t^r D^+_{\mathrm{dR}} \to 0,$$

where

$$V_{\mathrm{st}}(D, D^+_{\mathrm{dR}}) := \mathrm{Ker}((D \otimes_{F^{\mathrm{nr}}} B_{\mathrm{st}})^{N=0,\varphi=1} \to (D \otimes_{F^{\mathrm{nr}}} B_{\mathrm{dR}})/D^+_{\mathrm{dR}}).$$

Here we take $(D, D^+_{\mathrm{dR}}) := (H^i_{\mathrm{HK}}(U), H^i_{\text{ét}}(U, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B^+_{\mathrm{dR}})$. Then for $i \leq r$ we have

$$D \otimes_{F^{\mathrm{nr}}} B^+_{\mathrm{dR}} \simeq H^i_{\mathrm{dR}}(U/B^+_{\mathrm{dR}}), \quad t^r D^+_{\mathrm{dR}} = \mathrm{Fil}^r(H^i_{\text{ét}}(U, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{dR}}) \simeq \mathrm{Fil}^r H^i_{B^+_{\mathrm{dR}}}(X),$$

where the last isomorphism follows from Theorem **??**, and

$$(D \otimes_{F^{\mathrm{nr}}} B^+_{\mathrm{dR}})/t^r D^+_{\mathrm{dR}} \simeq H^i_{\mathrm{dR}}(U/B^+_{\mathrm{dR}})/t^r D^+_{\mathrm{dR}} \simeq H^i_{B^+_{\mathrm{dR}}}(X)/\mathrm{Fil}^r,$$

where the last isomorphism follows from [**?**, Proposition 5.17]. Using the semistable comparison, the geometric log de Rham comparison (Theorem **??**), and the fundamental exact sequence, we have

$$V_{\mathrm{st}}(D, D^+_{\mathrm{dR}}) \simeq H^i_{\text{ét}}(U, \mathbb{Q}_p),$$

and (D, D^+_{dR}) is a weakly admissible filtered (φ, N)-module over C (see [**?**, Remark 5.14]). The φ-slopes of $H^i_{\mathrm{HK}}(U)$ are in $[0, i]$, since this is true for quasi-compact dagger varieties, and we have the Gysin sequence. Then for $i \leq r$ the above theorem shows that we have the following short exact sequence:

$$0 \to H^i_{\text{ét}}(U, \mathbb{Q}_p(r)) \to (H^i_{\mathrm{HK}}(U) \otimes_{F^{\mathrm{nr}}} B^+_{\mathrm{st}})^{N=0,\varphi=p^r} \to H^i_{B^+_{\mathrm{dR}}}(X)/\mathrm{Fil}^r \to 0.$$

(5) The short exact sequence is Galois equivariant. Suppose that X, Z, U descend to X_0, Z_0, U_0 over K, respectively. We need to show that the morphism $\alpha_{\mathrm{st}}(U) \otimes_{B_{\mathrm{st}}} B_{\mathrm{dR}}$ is the same as the period morphism in Theorem **??**. Suppose first that Z_0 is smooth. After tensoring the commutative diagram (**??**) with B_{dR} over B_{st}, we obtain the commutative diagram

$$\begin{array}{ccccc}
\mathrm{R}\Gamma_{\text{ét}}(Z, \mathbb{Q}_p(-1))[-2] \otimes_{\mathbb{Q}_p} B_{\mathrm{dR}} & \longrightarrow & \mathrm{R}\Gamma_{\text{ét}}(X, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{dR}} & \longrightarrow & \mathrm{R}\Gamma_{\text{ét}}(U, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} B_{\mathrm{dR}} \\
\downarrow{\scriptstyle \alpha_{\mathrm{st}}(Z)(-1)[-2] \otimes_{B_{\mathrm{st}}} B_{\mathrm{dR}}} & & \downarrow{\scriptstyle \alpha_{\mathrm{st}}(X) \otimes_{B_{\mathrm{st}}} B_{\mathrm{dR}}} & & \downarrow{\scriptstyle \alpha_{\mathrm{st}}(U) \otimes_{B_{\mathrm{st}}} B_{\mathrm{dR}}} \\
\mathrm{R}\Gamma_{\mathrm{dR}}(Z_0)[-2] \otimes_K B_{\mathrm{dR}} & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{dR}}(X_0) \otimes_K B_{\mathrm{dR}} & \longrightarrow & \mathrm{R}\Gamma_{\mathrm{dR}}(U_0) \otimes_K B_{\mathrm{dR}}.
\end{array}$$

Since the period morphism in Theorem **??** also makes the above diagram commute by [**?**, Proposition 4.3.17], it suffices to check that, for proper X (and Z), $\alpha_{\mathrm{st}}(X) \otimes_{B_{\mathrm{st}}} B_{\mathrm{dR}}$ coincides with the period morphism constructed in [**?**], which follows from the local computation of Sally Gilles in [**?**]. In general, the same argument as in step (3) allows us to reduce the number of irreducible divisors in X. □

Remark 5.12. While working on the GAGA problem for Hyodo-Kato cohomology, the author observed that the semistable conjecture for U can also be deduced from the proper case and obtained the above proof. However, the proof intertwines different constructions of period morphisms and depends on Poincaré duality, which itself presents significant challenges. As mentioned in the introduction, a forthcoming article [**?**] will provide a more conceptual approach by introducing logarithmic syntomic cohomology and extending cohomology groups to the category of vector spaces. This perspective aligns with the methodologies developed in [**?**] and [**?**].

5.3. **Comparison with Tsuji's compactly supported log-crystalline cohomology.** In this section, we prove that the compactly supported Hyodo-Kato cohomology defined above agrees with the one previously defined by Tsuji in [**?**]. We begin by recalling Tsuji's construction.

5.3.1. *Definitions.* We start with the geometry of monoids, following [**?**] and [**?**].

Let P be a monoid. An ideal I of P is a subset of P satisfying $PI \subseteq I$. A prime ideal $\mathfrak{p}$ of P is an ideal of P such that $P - \mathfrak{p}$ is a submonoid of P. Let $\mathrm{Spec}(P)$ be the set of all primes of P. The dimension $\dim(P)$ of P is the maximal length r (if it exists) of a sequence $\mathfrak{p}_0 \supsetneq \mathfrak{p}_1 \supsetneq \cdots \supsetneq \mathfrak{p}_r$ (if no maximum exists, we define $\dim(P) = \infty$). The height $\mathrm{ht}(\mathfrak{p})$ of a prime ideal $\mathfrak{p}$ is the maximal length r (if it exists) of a sequence $\mathfrak{p} = \mathfrak{p}_0 \supsetneq \mathfrak{p}_1 \supsetneq \cdots \supsetneq \mathfrak{p}_r$ (if no maximum exists, we define $\mathrm{ht}(\mathfrak{p}) = \infty$).

For $h : Q \to P$ a morphism of monoids, we say a prime $\mathfrak{p}$ of P is horizontal with respect to h if $h(Q) \subset P - \mathfrak{p}$. For a morphism $f : (X, \mathcal{M}) \to (S, \mathcal{N})$ of log schemes and $x \in X$, we say a prime $\mathfrak{p}$ of $\mathcal{M}_{\overline{x}}$ is horizontal with respect to f if it is horizontal with respect to $f^*_{\overline{x}} : \mathcal{N}_{\overline{f(x)}} \to \mathcal{M}_{\overline{x}}$.

Let $f : (X, \mathcal{M}) \to (S, \mathcal{N})$ be a log-smooth morphism of fs log schemes. Define the sheaf of ideals $\mathcal{I}_f$ of the sheaf of monoids $\mathcal{M}$ by

$$\Gamma(U, \mathcal{I}_f) := \{a \in \Gamma(U, \mathcal{M}) | a \in \mathfrak{p} \text{ for all } \mathfrak{p} \in \mathrm{Spec}(\mathcal{M}_{\overline{x}}) \text{ of height } 1 \\ \text{horizontal with respect to } f \text{ and all } x \in U\}.$$

The ideal sheaf $\mathcal{I}_f\mathcal{O}_X$ of $\mathcal{O}_X$ is quasi-coherent. If $f : (X, \mathcal{M}) \to (S, \mathcal{N})$ is a log-smooth morphism of fs log schemes with $(S, \mathcal{N})$ log-regular, then we also have

$$\Gamma(U, \mathcal{I}_f) = \{a \in \Gamma(U, \mathcal{M}) | a \notin \mathcal{O}^*_{X,\overline{x}} \text{ for all } x \in S_U\},$$

where $S_U := \{x \in U \text{ of codimension 1 such that } \mathcal{M}_{\overline{x}} \neq \mathcal{O}^*_{X,\overline{x}} \text{ and } \mathcal{N}_{\overline{f(x)}} = \mathcal{O}^*_{S,\overline{f(x)}}\}$. In particular, we have the following description:

Proposition 5.13. *Let $\mathcal{I}$ be the ideal sheaf of $\mathcal{O}_X$ associated to the closed subset $\bigcup_{x \in S_X} \overline{\{x\}}$ of X with its reduced induced structure. Then $\mathcal{I} = \mathcal{I}_f\mathcal{O}_X$.*

Let $f : (X, \mathcal{M}) \to (S, \mathcal{N})$ be a log-smooth morphism of fs log schemes. Let $\Omega^\bullet_{X/S}$ be the associated complex of log-differentials. Then the compactly supported (log-)de Rham cohomology of X is defined by the complex

$$\mathrm{R}\Gamma_{\mathrm{dR,Tsu,c}}(X/S) := \mathrm{R}\Gamma(X, \mathcal{I}_f\mathcal{O}_X \otimes_{\mathcal{O}_X} \Omega^\bullet_{X/S}).$$

The compactly supported (log-)de Rham cohomology satisfies Poincaré duality.

Theorem 5.14. [**?**, Theorem 3.4] *Suppose moreover that X is proper and S is the log scheme $\mathrm{Spec}(\mathcal{O}_{F,n})$ equipped with the log structure induced by $\mathbb{N} \to \mathcal{O}_{F,n}, 1 \mapsto a$ for some $a \in \mathcal{O}_{F,n}$. Then*

(i) There is a natural trace map

$$\mathrm{tr}_{\mathrm{dR}} : \mathrm{R}\Gamma_{\mathrm{dR,Tsu,c}}(X/S)[2d] \to \mathcal{O}_{F,n}.$$

(ii) The pairing

$$H^i_{\mathrm{logdR}}(X/S) \otimes H^{2d-i}_{\mathrm{dR,Tsu,c}}(X/S) \to H^{2d}_{\mathrm{dR,Tsu,c}}(X/S) \to \mathcal{O}_{F,n}$$

is perfect.

We now review the definition of compactly supported Hyodo-Kato (log-crystalline) cohomology of Tsuji. Let $(S,\mathcal{N})$ (respectively $(S_n,\mathcal{N}_n)$) be the log scheme $\mathrm{Spec}(\mathcal{O}_F)$ (respectively $\mathrm{Spec}(\mathcal{O}_{F,n})$) equipped with the log structure induced by $\mathbb{N}\to\mathcal{O}_F, 1\mapsto 0$. Let $f:(X,\mathcal{M})\to(S_0,\mathcal{N}_0)$ be a log-smooth and universally saturated morphism of fs log schemes of relative dimension d. Denote by γ the canonical PD-structure on the ideal $p\mathcal{O}_S$. We will freely use the notation and properties of log-crystalline sites in [**?**].

We write $\mathcal{M}_{X/S}$ for the sheaf on the log-crystalline site $(X/S)_{\mathrm{cris}}$ given by

$$\Gamma((U,(T,\mathcal{M}_T)),\mathcal{M}_{X/S})=\Gamma(T,\mathcal{M}_T).$$

Denote by $u:(X/S)_{\mathrm{cris}}\to X_{\text{ét}}$ the canonical projection. Define the sheaf of ideals $\mathcal{I}_{X/S}$ of the sheaf of monoids $\mathcal{M}_{X/S}$ by

$$\begin{aligned}\Gamma((U,(T,\mathcal{M}_T)),\mathcal{I}_{X/S}):=\{a\in\Gamma(T,\mathcal{M}_T)|a\in\mathfrak{p}\text{ for all }x\in T\text{ and all }\mathfrak{p}\in\mathrm{Spec}(\mathcal{M}_{T,\overline{x}}/\mathcal{O}^*_{T,\overline{x}})\\ \text{of height 1 horizontal with respect to }\mathcal{N}_{\overline{f(x)}}/\mathcal{O}^*_{S,\overline{f(x)}}\to\mathcal{M}_{T,\overline{x}}/\mathcal{O}^*_{T,\overline{x}}\}.\end{aligned}$$

We write $\mathcal{K}_{X/S}$ for the ideal $\mathcal{I}_{X/S}\mathcal{O}_{X/S}$ of $\mathcal{O}_{X/S}$. It is a crystal by [**?**, Lemma 5.3]. In particular, if $(X,\mathcal{M})\to(Z,\mathcal{M}_Z)$ is a closed immersion into a log-smooth scheme over $(S_0,\mathcal{N}_0)$, and $(X^{\mathrm{PD}},\mathcal{M}^{\mathrm{PD}})$ is the PD-envelope of $(X,\mathcal{M})$ in $(Z,\mathcal{M}_Z)$, then there is a quasi-isomorphism

$$Ru_*\mathcal{K}_{X/S}\simeq\mathcal{I}_{f^{\mathrm{PD}}}\mathcal{O}_{X^{\mathrm{PD}}}\otimes_{\mathcal{O}_{X^{\mathrm{PD}}}}\Omega^\bullet_{X^{\mathrm{PD}}/S},$$

where f^{PD} is the map $(X^{\mathrm{PD}},\mathcal{M}^{\mathrm{PD}})\to(S_0,\mathcal{N}_0)$.

The compactly supported crystalline cohomology is defined by

$$\mathrm{R}\Gamma_{\mathrm{cris,Tsu,c}}(X/S):=R\varprojlim_n\mathrm{R}\Gamma((X/S_n)_{\mathrm{cris}},\mathcal{K}_{X/S_n})=R\varprojlim_n\mathrm{R}\Gamma_{\text{ét}}(X,Ru_*\mathcal{K}_{X/S_n}),$$

and the compactly supported Hyodo-Kato cohomology is then defined by

$$\mathrm{R}\Gamma_{\mathrm{HK,Tsu,c}}(X/S):=\mathrm{R}\Gamma_{\mathrm{cris,Tsu,c}}(X/S)\otimes_{\mathcal{O}_F}F.$$

Suppose that $(X,\mathcal{M})\to(Z,\mathcal{M}_Z)$ is a closed immersion into a log-smooth scheme over $(S_0,\mathcal{N}_0)$. Let $(X^{\mathrm{PD}},\mathcal{M}^{\mathrm{PD}})$ be its PD-envelope. Then we have a quasi-isomorphism

$$\mathrm{R}\Gamma_{\mathrm{cris,Tsu,c}}(X/S)\simeq\mathrm{R}\Gamma_{\mathrm{dR,Tsu,c}}(X^{\mathrm{PD}}/S).$$

Tsuji proved the following log-crystalline Poincaré duality theorem [**?**].

Theorem 5.15. [**?**, Proposition 5.5 and Theorem 5.6] *Suppose moreover that X is proper. Then*
(i) There is a natural trace map

$$\mathrm{tr}_{\mathrm{HK}}:\mathrm{R}\Gamma_{\mathrm{HK,Tsu,c}}(X/S)[2d]\to F.$$

(ii) The pairing

$$H^i_{\mathrm{HK}}(X)\otimes H^{2d-i}_{\mathrm{HK,Tsu,c}}(X/S)\to H^{2d}_{\mathrm{HK,Tsu,c}}(X/S)\to F$$

is perfect.

We have a description of Tsuji's compactly supported cohomology similar to that in the previous subsection.

Lemma 5.16. *Suppose $(U,\overline{U}) \in \mathrm{Var}_K^{\mathrm{ss}}$ is a strictly semistable pair as defined before Proposition **??**. Moreover, assume that $D := \overline{U}_\eta - U$ is a smooth irreducible divisor, and let $\mathscr{D} := \overline{D}$. Endow $(U,\overline{U})$ with the log structure defined by the compactifying log structure of the open immersion $U \hookrightarrow \overline{U}$. Then we have*

$$\mathrm{R}\Gamma_{\mathrm{HK,Tsu,c}}(\overline{U}_0) \simeq [\mathrm{R}\Gamma_{\mathrm{HK}}(\overline{U}_0) \to \mathrm{R}\Gamma_{\mathrm{HK}}(\mathscr{D}_0)].$$

Proof. Write i for the closed immersion $\mathscr{D}_0 \to \overline{U}_0$. It suffices to show that we have a distinguished triangle

$$Ru_*\mathcal{K}_{\overline{U}_0/S} \to Ru_*\mathcal{O}_{\overline{U}_0/S} \to Ru_*(i_*\mathcal{O}_{\mathscr{D}_0/S}).$$

Let $(\overline{U}_0^{\mathrm{PD}}, \mathcal{M}^{\mathrm{PD}})$ be the log PD-envelope of $(U,\overline{U})_0$, and let $(\mathscr{D}_0^{\mathrm{PD}}, \mathcal{M}_{\mathscr{D}_0}^{\mathrm{PD}})$ be the log PD-envelope of the closed immersion $\mathscr{D}_0 \hookrightarrow \overline{U}_0$. Since $(U,\overline{U}_0)$ is log-smooth over $(S,\mathcal{N})$, the above triangle is represented by

$$\mathcal{I}_{f^{\mathrm{PD}}}\mathcal{O}_{\overline{U}_0^{\mathrm{PD}}} \otimes_{\mathcal{O}_{\overline{U}_0^{\mathrm{PD}}}} \Omega^\bullet_{\overline{U}_0^{\mathrm{PD}}/S} \to \Omega^\bullet_{\overline{U}_0^{\mathrm{PD}}/S} \to i_*\Omega^\bullet_{\mathscr{D}_0^{\mathrm{PD}}/S}.$$

Since the log structure of $\overline{U}_0^{\mathrm{PD}}$ is induced by $\mathscr{D}_0^{\mathrm{PD}}$, the sheaf $\mathcal{I}_{f^{\mathrm{PD}}}\mathcal{O}_{\overline{U}_0^{\mathrm{PD}}}$ corresponds to the closed immersion of $\mathscr{D}_0 \hookrightarrow \overline{U}_0$. This concludes the proof. □

5.3.2. *Comparison with compactly supported algebraic Hyodo-Kato cohomology.*

Theorem 5.17. *Suppose that $(U,\overline{U}) \in \mathrm{Var}_K^{\mathrm{ss}}$ is a strictly semistable pair as defined before Proposition **??**. Then we have a quasi-isomorphism:*

$$\mathrm{R}\Gamma_{\mathrm{HK,Tsu,c}}(\overline{U}_0) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK,c}}(U),$$

which is compatible with Frobenius, monodromy, and the Hyodo-Kato isomorphism.

Proof. Using Poincaré duality (Theorems **??** and **??**), the quasi-isomorphism is constructed by the following dashed arrow:

$$\begin{array}{ccc} \mathrm{R}\Gamma_{\mathrm{HK,Tsu,c}}(\overline{U}_0) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK}}(U)^\vee \\ \vdots & & \downarrow\simeq \\ \mathrm{R}\Gamma_{\mathrm{HK,c}}(U) & \xrightarrow{\simeq} & \mathrm{R}\Gamma_{\mathrm{HK}}(U)^\vee. \end{array}$$

□

5.3.3. *Comparison with compactly supported analytic Hyodo-Kato cohomology.* We keep most of the notation above, but now let $\mathscr{X}$ be a proper semistable formal scheme over $\mathrm{Spf}(\mathcal{O}_K)$. Locally, $\mathscr{X}$ can be written as $\mathrm{Spf}(R)$ with R the completion of an étale algebra over

$$\mathcal{O}_K\langle x_1, x_2, ..., x_d, \frac{1}{x_1...x_a}, \frac{1}{x_{a+1}...x_{a+b}}\rangle,$$

and we equip $\mathscr{X}$ with the log structure $\mathscr{M}$ induced by the divisor $\mathscr{D} := (x_{a+1}...x_d = 0)$. Let Y be the special fiber $\mathscr{X}_0$ of $\mathscr{X}$. Denote by $\mathcal{M}_Y$ the log structure on Y induced by $\mathscr{M}$. Let X be the

rigid analytic variety over K associated to $\mathscr{X}$ and denote by X_{tr} its trivial locus. Locally, X_{tr} is equal to

$$\mathrm{Sp}(R\left[\frac{1}{p}\right]) \setminus (x_{a+1}...x_d = 0).$$

Since we also have Poincaré duality for almost proper rigid analytic varieties (Theorem **??**), the same argument as in the last subsection gives us the comparison with compactly supported analytic Hyodo-Kato cohomology.

Theorem 5.18. *We have a quasi-isomorphism:*

$$\mathrm{R}\Gamma_{\mathrm{HK,Tsu,c}}(\mathscr{X}_0) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{HK,c}}(X_{\mathrm{tr}}),$$

which is compatible with Frobenius, monodromy, and the Hyodo-Kato isomorphism.

Proof. This follows from Theorems **??** and **??**, using the same construction as in Theorem **??**. □

Appendix A. de Rham cohomology of Elmar Grosse-Klönne

In this appendix, we review the de Rham cohomology introduced by Elmar Grosse-Klönne in [**?**]. He defined de Rham cohomology for a rigid analytic variety X by locally embedding X into a smooth rigid analytic variety with a dagger structure. Here we restrict to dagger varieties. We show that this definition agrees with the overconvergent de Rham cohomology defined in [**?**] and [**?**]. We use this definition to prove the open-closed long exact sequence for cohomology with compact support in Proposition **??**.

Let X be a dagger variety over L, let Y be a smooth dagger variety over L, and let $\phi : X \hookrightarrow Y$ be a closed immersion. Such varieties X form a site $\mathrm{EmbRig}^{\dagger}_{L,\mathrm{an}}$ equipped with the analytic topology, by the following lemma. In fact, the lemma allows us to equip $\mathrm{EmbRig}^{\dagger}_{L}$ with any topology induced from $\mathrm{Rig}^{\dagger}_{L}$.

Lemma A.1. *Let $X, X', X'' \in \mathrm{EmbRig}^{\dagger}_{L}$, and let $f : X' \to X$ and $g : X'' \to X$ be morphisms. Then $X' \times_X X'' \in \mathrm{EmbRig}^{\dagger}_{L}$.*

Proof. Choose closed immersions $X \hookrightarrow Z$, $X' \hookrightarrow Z'$, and $X'' \hookrightarrow Z''$ with Z, Z', Z'' smooth. The graph constructions give closed immersions $X' \hookrightarrow Z \times Z'$ and $X'' \hookrightarrow Z \times Z''$. We therefore obtain a closed immersion $X' \times_X X'' = X' \times_Z X'' \hookrightarrow (Z \times Z') \times_Z (Z \times Z'') = Z \times Z' \times Z''$. □

It is clear that $\mathrm{EmbRig}^{\dagger}_{L}$ forms a basis of $\mathrm{Rig}^{\dagger}_{L,\mathrm{an}}$. We define a sheaf

$$\mathscr{A}_{\mathrm{dR,GK}} : (\mathrm{Rig}^{\dagger}_{L,\mathrm{an}})^{\mathrm{op}} \to \mathcal{D}(\mathrm{Mod}^{\mathrm{cond}}_{L})$$

as follows. For $X \in \mathrm{EmbRig}^{\dagger}_{L}$, let $\phi : X \hookrightarrow Y$ be a closed immersion into a smooth dagger variety Y. We denote by $\Phi(\phi, Y)$ the set of admissible open neighborhoods $U \subset Y$ of X, and by j_U the open immersion $U \to Y$. For an abelian sheaf $\mathcal{F}$ on Y, we define

$$\mathcal{F}^{\phi} = \mathrm{colim}_{U \in \Phi(\phi, Y)} j_{U,*}\mathcal{F}|_U.$$

We define $\mathscr{A}_{\mathrm{dR,GK}}$ to be the sheaf associated to the presheaf given by

$$X \mapsto \mathrm{R}\Gamma(X, \phi^{-1}(\Omega^{\bullet}_{Y/L})) = \mathrm{R}\Gamma(Y, (\Omega^{\bullet}_{Y/L})^{\phi}).$$

The following proposition guarantees that this definition makes sense.

Proposition A.2 ([**?**, Proposition 1.6])**.** *The complex* $\mathrm{R}\Gamma(X, \phi^{-1}(\Omega^\bullet_{Y/L}))$ *is independent of* Y *and* ϕ*, and depends only on the reduced structure of* X*. Moreover, it is contravariantly functorial in* X*.*

For $X \in \mathrm{Rig}^\dagger_L$, we define its de Rham cohomology in the sense of Grosse-Klönne by

$$\mathrm{R}\Gamma^{\mathrm{GK}}_{\mathrm{dR}}(X) := \mathrm{R}\Gamma(X_{\mathrm{an}}, \mathscr{A}_{\mathrm{dR,GK}}).$$

We have local-global compatibility:

Proposition A.3. *If* $X \in \mathrm{EmbRig}^\dagger_L$ *and* $\phi : X \hookrightarrow Y$ *is a closed immersion into a smooth dagger variety* Y*, then*

$$\mathrm{R}\Gamma^{\mathrm{GK}}_{\mathrm{dR}}(X) = R\Gamma(X, \phi^{-1}(\Omega^\bullet_{Y/L})).$$

Proof. The proof is similar to that of [**?**, 2.1]. Consider an embedded open covering $\mathcal{U} := \{X_i, Y_i\}_{i\in I}$, where

$$X^1 := \coprod_{i\in I} X_i \to X, \qquad X_i \hookrightarrow Y_i,$$

and each $X_i \hookrightarrow Y_i$ is a closed immersion into a smooth dagger variety Y_i. Set

$$X^n := (X^1)^{\times_X n} = \coprod_{(i)\in I^n} X_{(i)}, \qquad X_{(i)} \hookrightarrow Y_{(i)},$$

where, for $(i) = (i_1, \ldots, i_n) \in I^n$, we have $X_{(i)} = X_{i_1} \times_X \cdots \times_X X_{i_n}$, and the maps $\phi_{(i)} : X_{(i)} \hookrightarrow Y_{(i)} := Y_{i_1} \times \cdots \times Y_{i_n}$ are induced by the given embeddings. Thus we obtain simplicial objects $X^\bullet$ and $Y^\bullet$. For any $(i) \in I^n$, define the complex of sheaves $C_{(i)}$ on X by $C_{(i)} := j_{(i),*}\phi^{-1}_{(i)}\Omega^\bullet_{Y_{(i)}/L}$, where $j_{(i)} : X_{(i)} \to X$ is the open immersion. Define the Čech complex $\mathcal{C}(\mathcal{U})$ associated to $\{X_i, Y_i\}_{i\in I}$ by $\mathcal{C}^n(\mathcal{U}) = \prod_{(i)\in I^n} C_{(i)}$. The boundary maps are induced by the natural projections between the $Y_{(i)}$.

We need to prove that

$$R\Gamma(X, \phi^{-1}(\Omega^\bullet_{Y/L})) \simeq R\Gamma(X^\bullet, \phi^{-1}(\Omega^\bullet_{Y^\bullet/L})) = R\Gamma(X, \mathcal{C}(\mathcal{U})),$$

where the last equality follows from cohomological descent. It suffices to show that $R\Gamma(X, \mathcal{C}(\mathcal{U}))$ is independent of the embedded covering $\mathcal{U} := \{X_i, Y_i\}_{i\in I}$. For two embedded coverings $\mathcal{U} := \{X_i, Y_i\}_{i\in I}$ and $\mathcal{U}' := \{X'_j, Y'_j\}_{j\in J}$, we call $\mathcal{U}'$ a refinement of $\mathcal{U}$ if there is a map $\lambda : J \to I$ such that each X'_j is an open subset of $X_{\lambda(j)}$ and there is a smooth morphism $Y'_j \to Y_{\lambda(j)}$ compatible with the inclusion. For a refinement $\mathcal{U}'$ of $\mathcal{U}$, we have a natural map

$$u : \mathcal{C}(\mathcal{U}) \to \mathcal{C}(\mathcal{U}').$$

Since any two systems of local embeddings have a common refinement, it remains to show that $\mathrm{R}\Gamma(X, u)$ is an isomorphism.

The problem is local, so we may assume $X_1 = X'_1 = X$. Denote by $\mathcal{V}$ and $\mathcal{V}'$ the embedded open coverings $\{X_1, Y_1\}$ and $\{X'_1, Y'_1\}$, respectively. By the preceding proposition, we have

$$R\Gamma(X, \mathcal{C}(\mathcal{V})) \simeq R\Gamma(X, \mathcal{C}(\mathcal{V}')).$$

Since we have the commutative diagram

$$\begin{array}{ccc} \mathcal{C}(\mathcal{V}) & \longrightarrow & \mathcal{C}(\mathcal{V}') \\ \downarrow & & \downarrow \\ \mathcal{C}(\mathcal{U}) & \longrightarrow & \mathcal{C}(\mathcal{U}'), \end{array}$$

it suffices to show that the map $\mathcal{C}(\mathcal{V}) \to \mathcal{C}(\mathcal{U})$ is a quasi-isomorphism.

Write

$$\mathcal{C}(\mathcal{U}) = \mathcal{C}'(\mathcal{U}) \oplus \mathcal{C}''(\mathcal{U}),$$

where, for each n,

$$\mathcal{C}'^n(\mathcal{U}) := \prod_{(i)\in I^n, i_1 \neq 1} C_{(i)}, \qquad \mathcal{C}''^n(\mathcal{U}) := \prod_{(i)\in I^n, i_1 = 1} C_{(i)}.$$

For each $(i) = (i_1, \ldots, i_n) \in I^n$, the boundary map $\delta : C_{(i)} \to C_{(1,i_1,\ldots,i_n)}$ is an isomorphism, since $X_{(i)}$ and $X_{(1,i_1,\ldots,i_n)}$ are equal. Hence δ gives a quasi-isomorphism of $\mathcal{C}'(\mathcal{U})$ onto its image in $\mathcal{C}''(\mathcal{U})$, which consists of all terms except those of $\mathcal{C}(\mathcal{V})$. Thus the map $\mathcal{C}(\mathcal{V}) \to \mathcal{C}(\mathcal{U})$ is a quasi-isomorphism. This concludes the proof. □

Remark A.4. We consider only the overconvergent setting. The definition of $\mathrm{R}\Gamma_{\mathrm{dR}}^{\mathrm{GK}}(X)$ in this article agrees with the definition of $\mathrm{R}\Gamma_{\mathrm{dR}}(\widehat{X})$ in [**?**], so all propositions in [**?**] apply here.

The main result of this appendix is the following theorem.

Theorem A.5. *For $X \in \mathrm{Rig}_L^\dagger$, there is a natural isomorphism in $D(\mathrm{Mod}_L^{\mathrm{cond}})$:*

$$\mathrm{R}\Gamma_{\mathrm{dR}}^{\mathrm{GK}}(X) \xrightarrow{\simeq} \mathrm{R}\Gamma_{\mathrm{dR}}(X).$$

Proof. The assertion is local, so we may assume that $X \in \mathrm{EmbRig}_L^\dagger$ and that $\phi : X \hookrightarrow Y$ is a closed immersion into a quasi-compact smooth dagger variety Y.

We construct the map as the composition

$$\mathrm{R}\Gamma_{\mathrm{dR}}^{\mathrm{GK}}(X) = \mathrm{R}\Gamma(X, \phi^{-1}(\Omega^\bullet_{Y/L})) \to \mathrm{R}\Gamma(X, \Omega^\bullet_{X/L}) \to \mathrm{R}\Gamma_{\mathrm{dR}}(X).$$

The first map is canonical. To construct the second map, let $\pi : \mathrm{Rig}_{L,\text{éh}}^\dagger \to \mathrm{Rig}_{L,\text{an}}^\dagger$ be the natural map between big sites. Recall that $\Omega^i_{\text{éh}}$ is the éh-sheafification of the sheaf of continuous differentials, i.e., $\Omega^i_{\text{éh}} = \pi^{-1}\Omega^i_{/L}$. The second map is induced by the unit of the adjunction:

$$\Omega^\bullet_{/L} \to R\pi_*\pi^{-1}\Omega^\bullet_{/L} = R\pi_*\Omega^\bullet_{\text{éh}}.$$

We proceed by induction on the dimension of X. By [**?**, Proposition 2.1] and Proposition **??**, both sides satisfy the Mayer-Vietoris property for closed covers. Induction on the number of irreducible components therefore reduces us to the case where X is irreducible. If X is smooth, then by [**?**, Proposition 1.8] and Theorem **??**, both sides agree with the usual de Rham cohomology defined by continuous differentials, and the map constructed above is an isomorphism. In general, we resolve singularities and use [**?**, Proposition 2.2] and Proposition **??** to reduce to the smooth case. □

The proof of the theorem relies on embedded desingularization for dagger varieties, which follows from [**?**].

Theorem A.6 (Embedded desingularization)**.** *Let X be a quasi-compact smooth dagger variety over L, and let $Y \hookrightarrow X$ be a closed immersion. Then there exists a finite sequence of blowups $X' = X_n \to X_{n-1} \to \cdots \to X_0 = X$ such that the strict transform of Y is smooth and the center of each blowup is smooth and contained in the preimage of Y.*

Proof. Every dagger L-algebra is excellent. It suffices to prove this for the Washnitzer algebra W_n. By [**?**, Proposition 1.5], W_n is regular and equidimensional of dimension n, and it satisfies the conditions of [**?**, Theorem 102]. The argument in [**?**, Section 5] then yields the analogue of [**?**, Theorem 1.1.9] for dagger varieties. □

We consequently obtain the following finiteness result in our setting.

Proposition A.7. *Let X be a quasi-compact smooth dagger variety over $L = K$ or C, let $U \subset X$ be a quasi-compact open subset, let $Z \hookrightarrow X$ be a closed immersion, and set $T = X - (U \cup Z)$. For $i \geq 0$, the condensed cohomology group $H^i_{\mathrm{HK}}(T)$ is a finite-dimensional condensed vector space over L_F, where $L_F = F$ if $L = K$ and $L_F = F^{\mathrm{nr}}$ if $L = C$. Likewise, $H^i_{\mathrm{dR}}(T)$ is a finite-dimensional condensed vector space over L.*

Proof. When X is defined over C, the proposition follows from the Hyodo-Kato isomorphism together with [**?**, Corollary 3.5 and Theorem 3.6]; it remains to pass to the framework of condensed mathematics. It suffices to do this for de Rham cohomology. As in [**?**], we reduce to two cases: $X - U$, where X is a smooth dagger affinoid variety and U is a rational subdomain of X; and $X - Z$, where X is a smooth dagger affinoid variety and Z is a divisor with strict normal crossings. The first case follows from [**?**, Theorem 3.6] and the fact that $\mathrm{R}\Gamma_{\mathrm{dR}}(X - U)$ can be written as a filtered colimit of de Rham cohomology complexes of Stein spaces with finite-dimensional de Rham cohomology groups. The second case follows from [**?**, Corollary 3.5], [**?**, Theorem 2.8], and [**?**, Proposition 2.10]. The result over K for Hyodo-Kato cohomology then follows by Galois descent. □

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